

OPTIMAL FEEDBACK LAW RECOVERY BY GRADIENT-AUGMENTED NEURAL NETWORK WITH NONCONVEX REGULARIZATION ON THE UNBOUNDED PARAMETER SPACE

ABSTRACT. We recover optimal feedback laws for nonlinear control problems by approximating Hamilton–Jacobi–Bellman value functions with sparse shallow neural networks. Value and gradient are generated from the open-loop problem by the Pontryagin maximum principle, and the network is trained with a gradient-augmented H^1 fidelity.

For nonhomogeneous activation functions on the unbounded parameter space $\Omega = \mathbb{R}^{d+1}$, We study the normalized objective

$$J(\mu) = L(\mu) + \alpha \Phi_\phi(\mu_p),$$

where the concave scalar penalty Φ_ϕ has finite positive right slope at zero, and $\mu_p = w_p \mu$ with $p > 0$ and $w_p(\omega) = 1 + |\omega|^p$. Normalization makes the dictionary vanish at infinity under a polynomial growth condition. This yields existence of the global minimum by normalized weak-* compactness. If the penalty is strongly concave on every bounded interval, every local minimizer has finite support and satisfies a covering-number bound. A Primal-Dual-Active-Point (PDPA) algorithm is proposed, which achieves a best-iterate rate of order $n^{-1/2}$.

For positively k -homogeneous activations, we give an equivalent measure problem on the unit sphere with the fractional outer-coefficient penalty $|c|^{2/(k+1)}$. Every local minimizer has finite support, while every global minimizer satisfies an explicit a priori width bound. An insertion algorithm based on the optimality condition is proposed.

The resulting algorithms are studied on the Van der Pol oscillator and the pendulum swing-up problems, whose value function has a gradient discontinuity across a switching set.

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1. INTRODUCTION

The problem under consideration is the dynamic optimal control problem in the space-time domain

$$(P) \quad \begin{aligned} & \inf_{u \in L^2(t_0, T; \mathbb{R}^{m_u})} \mathcal{C}(u; t_0, x) := \int_{t_0}^T (h(y(t)) + \eta |u(t)|^2) dt, \quad \eta > 0, \\ & \text{s.t.} \\ & \dot{y} = f(y) + g(y)u, \\ & y(t_0) = x, \end{aligned}$$

We assume $y(t) \in \mathbb{R}^d$, $u(t) \in \mathbb{R}^{m_u}$, $h \in C^1(\mathbb{R}^d)$, $f \in C^1(\mathbb{R}^d; \mathbb{R}^d)$, and $g \in C^1(\mathbb{R}^d; \mathbb{R}^{d \times m_u})$. We assume the minimizer exists.

One approach computes an open-loop state-costate-control path for a fixed initial condition x . The Pontryagin maximum principle (PMP) yields a coupled forward-backward two-point boundary value problem:

$$(TPBVP) \quad \begin{cases} \frac{d}{dt} y^* = f(y^*) + g(y^*)u^*, \\ y^*(t_0) = x, \\ -\frac{d}{dt} p^* = \partial_y \left(f(y^*) + g(y^*)u^* \right)^\top p^* + \partial_y h(y^*), \\ p^*(T) = 0. \end{cases}$$

The stationarity condition with respect to the control gives

$$u^*(t) = -\frac{1}{2\eta} g(y^*(t))^\top p^*(t) \quad \text{for almost every } t \in (t_0, T).$$

The open-loop control is not robust with respect to disturbances of the state [1]: it is computed for a single initial condition, and deviations from the optimal path are not corrected. In addition, numerical methods for the two-point boundary value problem are sensitive to the initial guess and may fail to converge. The dynamic programming principle instead introduces the value function

$$V(t, x) := \inf_u \mathcal{C}(u; t, x)$$

which satisfies the Hamilton–Jacobi–Bellman (HJB) equation in the viscosity sense:

$$(HJB) \quad \begin{cases} \partial_t V - \frac{1}{4\eta} \nabla V(t, x)^\top g(x) g(x)^\top \nabla V(t, x) + \nabla V(t, x)^\top f(x) + h(x) = 0, \\ V(T, x) = 0. \end{cases}$$

At points where V is differentiable, the equation holds classically and Hamiltonian minimization gives the feedback $u^*(t, x) = -(2\eta)^{-1} g(x)^\top \nabla V(t, x)$. The value function encodes the optimal cost from every state. The feedback law synthesized from it corrects deviations from a nominal path.

For nonlinear control problems one usually seeks a feedback law, which requires solving the HJB equation. Computing this function is notoriously difficult because of the curse of dimensionality. Alongside classical methods, data-driven methods have therefore become increasingly common. One line of work, introduced in [18], generates the training data with (TPBVP) and approximates the optimal value function from it. The optimal feedback law is then synthesized from the fitted value function after validation. Its samples are causality-free: generating one does not require approximate values at neighbouring grid points. Moreover, Azmi, Kalise and Kunisch [1] enrich the dataset with adjoint information: a state-adjoint-control triple is generated by solving the open-loop problem, and at differentiability points the adjoint supplies the value gradient. The value function is then approximated by hyperbolic-cross basis functions with an ℓ^1 penalty, and

including the adjoint data targets an H^1 approximation. Neural-network approximations form another growing line of work [16, 23]. Other prior information, such as semiconcavity of the value function, can also be incorporated. Alternatively, one can use an HJB- or physics-informed loss [9, 17, 22, 24, 15].

For $\omega = (\omega_n)_{n=1}^N$, $\omega_n = (a_n, b_n)$, and $c = (c_n)_{n=1}^N$, we use the two-layer network

$$(1) \quad \mathcal{N}_{\omega,c}(x) := \sum_{n=1}^N c_n \rho(a_n \cdot x + b_n) \in \mathbb{R},$$

where ρ is a sufficiently smooth activation function and N the width of the network. We associate it with the finite signed measure $\mu_{\omega,c} := \sum_{n=1}^N c_n \delta_{\omega_n}$. Its gradient is

$$\nabla_x \mathcal{N}_{\omega,c}(x) = \sum_{n=1}^N c_n \nabla_x \rho(a_n \cdot x + b_n) \in \mathbb{R}^d.$$

We write $V(x) := V(0, x)$. The training data are generated by solving (TPBVP). For the trajectory issued from x^m , let

$$g^m := p^{*,m}(0)$$

denote the costate. At every sample where V is differentiable, sensitivity relations give $g^m = \nabla V(x^m)$ [8, 6, 7]. At a nondifferentiable point, competing optimal paths may supply different costate labels; no classical gradient is asserted there. Thus the data consist of triples $\{x^m, V(x^m), g^m\}_{m=1}^M$. We have the minimization problem

$$\min_{N, \omega, c} l^M(\mu_{\omega,c}), \quad l^M(\mu_{\omega,c}) := \frac{1}{2M} \sum_{m=1}^M (|V(x^m) - \mathcal{N}_{\omega,c}(x^m)|^2 + |g^m - \nabla \mathcal{N}_{\omega,c}(x^m)|^2).$$

Fix a bounded Lipschitz domain $D \subset \mathbb{R}^d$. Section 3 introduces the corresponding population fidelity L with respect to Lebesgue measure on D .

Another focus of this study is the sparsity of the network. The number of neurons N is itself an optimization variable. Although width alone does not generally explain a model's inductive bias [19], sparsity is crucial for computational efficiency in many problems. Furthermore, sparsity is necessary for certain applications, such as knowledge distillation and deconvolution of point-like objects.

A traditional sparse shallow network uses the ReLU activation and an ℓ^1 penalty on the outer coefficients [4, 2]. Pieper and Petrosyan [20] showed that convex penalties cannot effectively eliminate redundant neurons when they cluster in a small neighborhood and introduced a nonconvex penalty for this purpose. However this formulation does not extend to the gradient fitting since the formulation is restricted to the ReLU setting: the second-order weak derivative of ReLU does not exist as a function while it is a distribution, and we cannot use it to approximate high-order derivatives. We need to extend to the smooth activation functions.

The difficulty is, such extension requires a treatment of the unbounded inner-parameter space for gradient training. 1. For positively k -homogeneous activations, radial scaling can be absorbed into the outer coefficient, and the network has an equivalent representation with inner parameters on the unit sphere. However, the induced nonconvex penalty admits infinite slope at 0 and the usual functional tools to prove the existence and optimality are not applicable. 2. For nonhomogeneous activations such as tanh, softplus, and the Gaussian, radial scaling changes the atom and cannot be removed by normalization. Without normalization, sequences with $|\omega| \rightarrow \infty$ may retain a nonzero contribution to the represented function, as shown in Section 3.3. We address those difficulties in each case.

Contributions. The contributions are the following.

- (1) *Nonhomogeneous activations on the unbounded parameter domain.* We adopt the objective function with gradient information from [1] and the nonconvex penalty from [20]. Like in [20], we require the concave scalar penalties to have finite positive right slope. However, unlike the original paper which only considers the ReLU case, we allow the activation functions to be nonhomogeneous defined on the unbounded domain. We prove a weak representation of the formulation and derive weak-* lower semicontinuity (Theorem 3.9 and Corollary 3.10). In Theorem 3.13 with the normalized atom K_p , we extend the existence result of a global minimizer from the value loss with ReLU activation function to the gradient-augmented loss with any continuous activations on $\Omega = \mathbb{R}^{d+1}$ of implied polynomial growth condition (Theorem 3.16). Under strong concavity of the penalty on every bounded interval, we proved that every local minimizer is finite atomic as in [20]. However, we further extend the result and prove that the number of neurons of optimal network admits an a priori bound (Theorem 3.20, Corollary 3.21). Based on that, we propose a new version of the PDAP algorithm with $n^{-1/2}$ best-iterate rate (Theorem 5.1, Corollary 5.2).
- (2) *The k -homogeneous formulation.* For positively k -homogeneous activations, we show that quadratic regularization of the inner and outer parameters is equivalent to a penalty proportional to $|c|^{2/(k+1)}$ on the outer coefficients with unit-norm inner parameters (Lemma 4.3), which is extended from the ReLU rescaling equivalence of [20, Appendix F, Proposition 3]. The resulting problem is posed over all of $\mathcal{M}(\mathbb{S}^d)$, which has the same infimum as the finite-network problem (Proposition 4.6). We prove that a global minimizer exists, and every global minimizer is a finite network obeying an explicit a priori width bound (Theorem 4.9). We derive the atomwise optimality conditions and a positive lower bound on the weights of every total-variation local minimizer (Theorem 4.7). Every local minimizer has finite support (Corollary 4.11). An insertion algorithm is proposed based on the optimality condition (Algorithm 2).

Figure 1 shows that the proposed algorithms can achieve lower relative H^1 validation errors with fewer neurons than the traditional ℓ^1 -regularized ReLU network on both benchmarks.

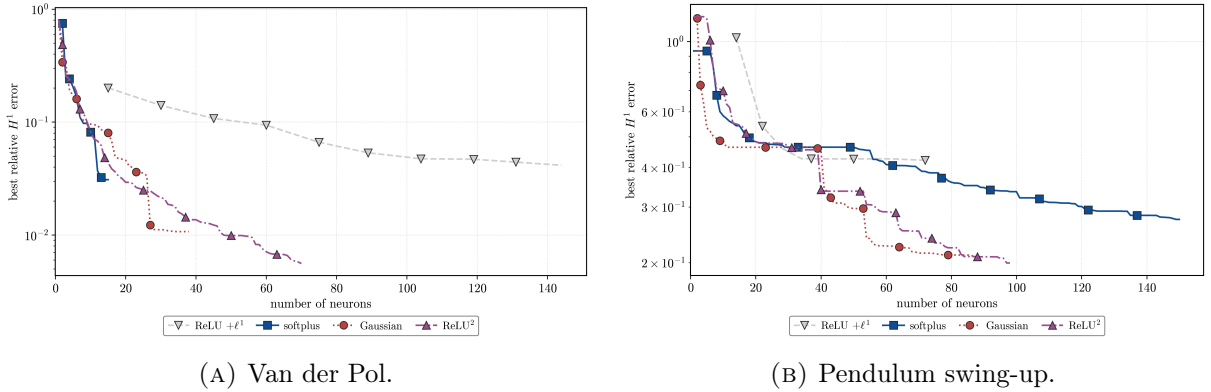


FIGURE 1. Best observed relative H^1 validation error at each support budget for (a) Van der Pol and (b) Pendulum swing-up. See Sections 6.1 and 6.2 for the experimental settings.

2. PRELIMINARIES

This section reviews the measure-theoretic facts used in the paper. The learning objectives and assumptions are introduced in Section 3.

2.1. Finite signed Radon measures. Let Ω be a locally compact Polish space. We write $\mathcal{M}(\Omega)$ for the Banach space of finite signed Radon measures on Ω , equipped with the total-variation norm

$$\|\mu\|_{\mathcal{M}(\Omega)} := |\mu|(\Omega).$$

By the Riesz representation theorem, $\mathcal{M}(\Omega) = C_0(\Omega)^*$ under the pairing

$$\langle f, \mu \rangle := \int_{\Omega} f d\mu.$$

Here $C_0(\Omega)$ denotes the continuous functions vanishing at infinity.

For $\mu \in \mathcal{M}(\Omega)$ we define the *atom set* $\text{atom } \mu := \{\omega \in \Omega : |\mu|(\{\omega\}) > 0\}$, which is at most countable, the purely atomic part $\mu_{\text{atom}} := \sum_{\omega \in \text{atom } \mu} \mu(\{\omega\}) \delta_{\omega}$, and the nonatomic part $\mu_{\text{cont}} := \mu - \mu_{\text{atom}}$ [5]. This gives the decomposition $\mu = \mu_{\text{cont}} + \mu_{\text{atom}}$. The atom set need not be closed because atoms may accumulate at a point that carries no mass. The polar decomposition of a signed measure is written as

$$\mu = \text{sign}(\mu) |\mu|, \quad \text{sign}(\mu) := \frac{d\mu}{d|\mu|} \in \{-1, 1\} \quad |\mu|\text{-almost everywhere.}$$

2.2. Measures with a finite moment. In this subsection let $\Omega = \mathbb{R}^{d_{\Omega}}$, where $d_{\Omega} \geq 1$. For $p > 0$, set

$$w_p(\omega) := 1 + |\omega|^p$$

and define

$$(2) \quad \mathcal{M}_p(\Omega) := \left\{ \mu \in \mathcal{M}(\Omega) : \int_{\Omega} w_p d|\mu| < \infty \right\}, \quad \|\mu\|_{\mathcal{M}_p(\Omega)} := \int_{\Omega} w_p d|\mu|.$$

The associated space of continuous functions is

$$C_{w_p^{-1}}(\Omega) := \left\{ f \in C(\Omega) : \frac{f}{w_p} \in C_0(\Omega) \right\}, \quad \|f\|_{C_{w_p^{-1}}} := \sup_{\omega \in \Omega} \frac{|f(\omega)|}{w_p(\omega)}.$$

Under the dual pairing, $\mathcal{M}_p(\Omega) = C_{w_p^{-1}}(\Omega)^*$ isometrically [3, Section 5].

Notation. Throughout the remainder of the paper, $D \subset \mathbb{R}^d$ is a bounded Lipschitz domain, $\nu := \mathcal{L}^d|_D$ is the population measure, and $L^2(D)$ means $L^2(D, \nu)$. We assume that the population target V belongs to $H^1(D)$. The symbol $M \geq 1$ denotes the number of samples in an empirical loss. The inner-parameter domain is $\Omega = \mathbb{R}^{d+1}$, and $\rho \in C(\mathbb{R})$ denotes the fixed activation function of (1), with

$$\omega = (a, b), \quad [K(\omega)](x) := \rho(a \cdot x + b).$$

With this notation, the network in (1) is $\mathcal{N}_{\omega, c} = \sum_{n=1}^N c_n K(\omega_n)$. The symbol δ_{ω} denotes the Dirac measure at ω . We reserve p for the moment order.

3. THE LEARNING PROBLEM

3.1. The value operator and the gradient-augmented loss. The finite network $\mathcal{N}_{\omega, c}$ corresponds to the finite atomic measure $\mu_{\omega, c}$. This motivates the integral formulation below, which agrees exactly with the finite sum on atomic measures.

Assumption 3.1 (Regularity and polynomial growth of ρ). The activation satisfies $\rho \in C^1(\mathbb{R})$. There are $C_{\rho} > 0$, $s_0 \geq 0$, and $s_1 \geq \max\{s_0, 1\}$ such that

- (1) $|\rho(z)| \leq C_{\rho}(1 + |z|)^{s_0}$ for every $z \in \mathbb{R}$;
- (2) $|\rho'(z)| \leq C_{\rho}(1 + |z|)^{s_1-1}$ for every $z \in \mathbb{R}$.

Since D is bounded, Assumption 3.1 gives constants $C_D, C_K > 0$ such that

$$(3) \quad \|K(\omega)\|_{L^2(D)} \leq C_D(1 + |\omega|)^{s_0}.$$

Moreover, $\nabla K(a, b)(x) = a\rho'(a \cdot x + b)$, and hence

$$(4) \quad \|K(\omega)\|_{H^1(D)} \leq C_K(1 + |\omega|)^{s_1}.$$

If $\omega_n \rightarrow \omega$, their pre-activations lie in a common compact interval on D . Uniform continuity of ρ and ρ' on that interval, together with $a_n \rightarrow a$, therefore gives $K(\omega_n) \rightarrow K(\omega)$ in $H^1(D)$. Thus $K : \Omega \rightarrow H^1(D)$ is continuous and, in particular, strongly measurable by Pettis.

For $\mu \in \mathcal{M}(\Omega)$ satisfying

$$(5) \quad \int_{\Omega} \|K(\omega)\|_{H^1(D)} d|\mu|(\omega) < \infty,$$

define the $H^1(D)$ -valued Bochner integral

$$\mathcal{N}\mu := \int_{\Omega} K(\omega) d\mu(\omega).$$

By (4), condition (5) holds for $\mu \in \mathcal{M}_p(\Omega)$ when $p > s_1$. For $\psi \in C_c^\infty(D)$ and $i \in \{1, \dots, d\}$, the growth bound permits Fubini's theorem and gives

$$\begin{aligned} \int_D (\mathcal{N}\mu) \partial_i \psi dx &= \int_{\Omega} \int_D K(\omega) \partial_i \psi dx d\mu(\omega) \\ &= - \int_D \left(\int_{\Omega} \partial_i K(\omega) d\mu(\omega) \right) \psi dx. \end{aligned}$$

Consequently,

$$\nabla(\mathcal{N}\mu) = \int_{\Omega} \nabla K(\omega) d\mu(\omega) \quad \text{in } L^2(D; \mathbb{R}^d).$$

For measures satisfying only $\int_{\Omega} \|K(\omega)\|_{L^2(D)} d|\mu|(\omega) < \infty$, the same definition is valid as an $L^2(D)$ -valued Bochner integral. By (3), this includes $\mathcal{M}_p(\Omega)$ when $p > s_0$. In particular, $\mathcal{N}\mu_{\omega, c} = \mathcal{N}_{\omega, c}$.

Whenever $\mathcal{N}\mu \in H^1(D)$, set $r_\mu := \mathcal{N}\mu - V$. The fidelity term is the gradient-augmented loss with the distributional gradient convention. If the $L^2(D)$ -valued integral defining $\mathcal{N}\mu$ does not exist, we set $L(\mu) = +\infty$:

$$L(\mu) := \begin{cases} \frac{1}{2} \|\mathcal{N}\mu - V\|_{L^2(D)}^2 + \frac{1}{2} \|\nabla(\mathcal{N}\mu) - \nabla V\|_{L^2(D)}^2, & \mathcal{N}\mu \in H^1(D), \\ +\infty, & \text{otherwise.} \end{cases}$$

For sampled labels $(V(x^m), g^m)_{m=1}^M$ and a finite network for which the displayed point values exist, the corresponding empirical fidelity is

$$(6) \quad l^M(\mu) := \frac{1}{2M} \sum_{m=1}^M (|\mathcal{N}\mu(x^m) - V(x^m)|^2 + |\nabla \mathcal{N}\mu(x^m) - g^m|^2).$$

3.2. The nonconvex functional. Let $\phi : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ satisfy the following assumptions.

Assumption 3.2. The finite right derivative

$$L_\phi := \phi'(0+)$$

exists and satisfies $0 < L_\phi < \infty$. We use the convention $\phi'(0) := L_\phi$.

Assumption 3.3. The function ϕ is concave, nondecreasing, and coercive, with $\phi(0) = 0$, and belongs to $C^1((0, \infty))$.

Assumptions 3.2 and 3.3 imply that $\phi'(t) > 0$ for every $t > 0$. Indeed, if $\phi'(t_0) = 0$ at some $t_0 > 0$, concavity and monotonicity would give $\phi'(t) = 0$ for every $t \geq t_0$, making ϕ constant there and contradicting coercivity. Hence ϕ is continuous and strictly increasing from $[0, \infty)$ onto $[0, \infty)$. We denote its inverse by ϕ^{-1} .

Assumption 3.4 (Local curvature). The function ϕ belongs to $C^2((0, \infty))$ and satisfies both of the following conditions:

- (1) there are $\gamma_1 \geq 0$ and $\hat{z}_1 > 0$ such that $-\gamma_1(z_2 - z_1) \leq \phi'(z_2) - \phi'(z_1) \leq 0$ for $0 \leq z_1 \leq z_2 \leq \hat{z}_1$;
- (2) for every $A > 0$,

$$\gamma_A := - \sup_{0 < t \leq A} \phi''(t) > 0.$$

Remark 3.5. The lower curvature control in Assumption 3.4(1) is local and does not follow from global concavity. For example, $\phi(z) = z - \frac{2}{3}z^{3/2}$ near the origin has $\phi''(z) = -\frac{1}{2}z^{-1/2}$; it can be extended linearly after a small positive point so as to remain nondecreasing, concave, and coercive.

The following result follows from standard concavity estimates; see [20].

Lemma 3.6. *Let $\phi : [0, \infty) \rightarrow [0, \infty)$ be concave with $\phi(0) = 0$. Then*

- (1) $\phi(z_1 + z_2) \leq \phi(z_1) + \phi(z_2)$ for all $z_1, z_2 \geq 0$.

If in addition $L_\phi := \phi'(0+)$ is finite, then

- (2) $\phi(z) \leq L_\phi z$ for all $z \geq 0$;
- (3) $\phi(z)/z \rightarrow L_\phi$ as $z \downarrow 0$;
- (4) for every nonnegative sequence $(t_j)_{j \geq 1}$ with $\sum_{j \geq 1} t_j < \infty$,

$$\phi\left(\sum_{j \geq 1} t_j\right) \leq \sum_{j \geq 1} \phi(t_j).$$

Proof. Concavity and $\phi(0) = 0$ make $z \mapsto \phi(z)/z$ nonincreasing on $(0, \infty)$: for $0 < z_1 \leq z_2$, writing $z_1 = \frac{z_1}{z_2}z_2 + (1 - \frac{z_1}{z_2}) \cdot 0$ and using concavity together with $\phi(0) = 0$ gives $\phi(z_1) \geq \frac{z_1}{z_2}\phi(z_2)$.

- (1) Suppose without loss of generality that $0 < z_1 \leq z_2$. Then

$$\begin{aligned} \phi(z_1 + z_2) &\leq \frac{z_1 + z_2}{z_2} \phi(z_2) && \text{because } \phi(z)/z \text{ is nonincreasing} \\ &= \phi(z_2) + z_1 \frac{\phi(z_2)}{z_2} \\ &\leq \phi(z_2) + \phi(z_1) && \text{by the same monotonicity.} \end{aligned}$$

The cases with $z_1 = 0$ or $z_2 = 0$ follow from $\phi(0) = 0$.

- (2) and (3) By the same monotonicity and $\phi(0) = 0$,

$$\frac{\phi(z)}{z} \leq \lim_{s \downarrow 0} \frac{\phi(s)}{s} = \lim_{s \downarrow 0} \frac{\phi(s) - \phi(0)}{s} = L_\phi \quad (z > 0),$$

and the displayed limit is precisely (3).

- (4) By (2) the function ϕ is continuous at 0, and a finite concave function is continuous on $(0, \infty)$. Iterating (1) over partial sums gives

$$\phi\left(\sum_{j \leq N} t_j\right) \leq \sum_{j \leq N} \phi(t_j) \leq \sum_{j \geq 1} \phi(t_j),$$

the last step because the terms are nonnegative. Letting $N \rightarrow \infty$ and using continuity of ϕ at $\sum_{j \geq 1} t_j$ proves the claim. $\square$

Assumption 3.3 implies the standing hypotheses of Lemma 3.6, and Assumption 3.2 supplies the finite right slope needed for parts (2)–(4). Neither coercivity nor differentiability on $(0, \infty)$ is used, so the lemma also applies under the weaker hypotheses of Theorem 3.9.

Proposition 3.7 (Second-order bounds). *Suppose that $\phi(0) = 0$ and Assumption 3.2 holds.*

- (1) *If the derivative bound in Assumption 3.4(1) holds, then $\phi(z) \geq L_\phi z - \frac{\gamma_1}{2} z^2$ for $0 \leq z \leq \hat{z}_1$;*
- (2) *if $\phi \in C^2((0, \infty))$ and Assumption 3.4(2) holds, then, for every $A > 0$, $\phi(z) \leq L_\phi z - \frac{\gamma_A}{2} z^2$ for $0 \leq z \leq A$.*

Proof. For $\varepsilon > 0$, apply the fundamental theorem of calculus on $[\varepsilon, z]$ and then let $\varepsilon \downarrow 0$. The definition of the finite right derivative and $\phi(0) = 0$ give

$$\phi(z) - L_\phi z = \int_0^z [\phi'(s) - L_\phi] ds.$$

The first conclusion follows from Assumption 3.4(1); see also [20].

For the second, Assumption 3.4(2) gives $\phi''(t) \leq -\gamma_A$ on $(0, A]$. Integrating this bound with respect to t from ε to s gives $\phi'(s) - \phi'(\varepsilon) \leq -\gamma_A(s - \varepsilon)$. Letting $\varepsilon \downarrow 0$ and using $\phi'(\varepsilon) \rightarrow L_\phi$, we obtain $\phi'(s) - L_\phi \leq -\gamma_A s$ for $0 < s \leq A$. Integrating this inequality with respect to s from 0 to z proves the claim. $\square$

For an open set $U \subset \Omega$, let $C_c(U)$ denote the continuous functions with compact support in U . Define

$$(7) \quad \Phi_\phi(\mu) := L_\phi \|\mu_{\text{cont}}\|_{\mathcal{M}(\Omega)} + \sum_{\omega \in \text{atom } \mu} \phi(|\mu(\{\omega\})|), \quad \mu \in \mathcal{M}(\Omega).$$

We give a self-contained representation and proof of weak-* lower semicontinuity for this specific nonconvex functional under relaxed assumptions. Another proof for the general setting can be found in Bouchitté–Buttazzo [5, Theorem 3.3 and Remark 3.5].

Lemma 3.8. *Let $U \subset \Omega = \mathbb{R}^{d+1}$ be open, let $\mu \in \mathcal{M}(U)$, and let $\delta > 0$. Suppose that*

$$|\mu(\{\omega\})| \leq \frac{\delta}{4} \quad \text{for every } \omega \in U.$$

Then, for every $\varepsilon > 0$, there are $m \geq 1$ and functions $h_1, \dots, h_m \in C_c(U)$ with pairwise disjoint supports and $|h_i| \leq 1$ such that

$$\left| \int_U h_i d\mu \right| \leq \delta \quad (i = 1, \dots, m), \quad \sum_{i=1}^m \left| \int_U h_i d\mu \right| \geq \|\mu\|_{\mathcal{M}(U)} - \varepsilon.$$

Proof. If $\varepsilon \geq \|\mu\|_{\mathcal{M}(U)}$, take $m = 1$ and $h_1 = 0$. Henceforth assume $0 < \varepsilon < \|\mu\|_{\mathcal{M}(U)}$. Let $U = P \cup N$ be a Hahn decomposition of μ . By inner regularity, there are disjoint compact sets $K_+ \subset P$ and $K_- \subset N$ such that

$$(8) \quad |\mu|(U \setminus (K_+ \cup K_-)) < \frac{\varepsilon}{4}.$$

We now construct the required finite small-mass partition, including both the atomic and nonatomic parts. For every $\omega \in K_+ \cup K_-$, continuity from above gives

$$|\mu|(B_r(\omega)) \longrightarrow |\mu|(\{\omega\}) \leq \frac{\delta}{4} \quad (r \downarrow 0).$$

Choose $r_\omega > 0$ such that $|\mu|(B_{r_\omega}(\omega)) < \delta/2$. Compactness gives a finite subcover. Let $B_1, \dots, B_N$ be these balls. For each $\sigma \in \{+, -\}$, define inductively

$$E_i^\sigma := (K_\sigma \cap B_i) \setminus \bigcup_{j < i} (K_\sigma \cap B_j), \quad i = 1, \dots, N,$$

where the union is empty for $i = 1$. After discarding empty sets, the E_i^σ form a pairwise disjoint Borel partition of K_σ , with $E_i^\sigma \subset B_i$. A finite Borel partition $E_1, \dots, E_{m_0}$ for some $m_0 \geq 1$ of $K_+ \cup K_-$ is constructed such that each E_i lies in either P or N and $|\mu|(E_i) < \delta/2$.

By inner regularity, choose compact sets $C_i \subset E_i$ whose total omitted mass is less than $\varepsilon/4$, and discard the empty sets. At least one set remains because $\varepsilon < \|\mu\|_{\mathcal{M}(U)}$ and (8). Relabel the remaining sets $C_1, \dots, C_m$. They are pairwise disjoint and satisfy

$$(9) \quad \sum_{i=1}^m |\mu|(C_i) \geq |\mu|(K_+ \cup K_-) - \frac{\varepsilon}{4}, \quad |\mu|(C_i) \leq \frac{\delta}{2}.$$

Each C_i lies entirely in either P or N .

Because the compact sets C_i are finitely many, pairwise disjoint, and contained in the open set U , they have pairwise disjoint relatively compact open neighborhoods. Outer regularity lets us shrink these neighborhoods, while keeping each C_i inside, to obtain open sets U_i satisfying

$$(10) \quad \sum_{i=1}^m |\mu|(U_i \setminus C_i) < \min \left\{ \frac{\varepsilon}{4}, \frac{\delta}{2} \right\}.$$

For each i , the continuous function

$$\chi_i(\omega) := \frac{\text{dist}(\omega, U_i^c)}{\text{dist}(\omega, C_i) + \text{dist}(\omega, U_i^c)}$$

equals one on C_i , vanishes outside U_i , and has compact support in U . Set $h_i = \chi_i$ when $C_i \subset P$ and $h_i = -\chi_i$ when $C_i \subset N$. Their supports are pairwise disjoint and $|h_i| \leq 1$. Moreover,

$$\left| \int_U h_i d\mu \right| \leq |\mu|(U_i) \leq |\mu|(C_i) + |\mu|(U_i \setminus C_i) \leq \delta, \quad \text{by (9)–(10),}$$

and

$$\begin{aligned} \sum_{i=1}^m \left| \int_U h_i d\mu \right| &\geq \sum_{i=1}^m (|\mu|(C_i) - |\mu|(U_i \setminus C_i)) \\ &\geq |\mu|(K_+ \cup K_-) - \frac{\varepsilon}{4} - \frac{\varepsilon}{4} && \text{by (9)–(10),} \\ &\geq \|\mu\|_{\mathcal{M}(U)} - \frac{\varepsilon}{4} - \frac{\varepsilon}{4} - \frac{\varepsilon}{4} && \text{by (8),} \\ &= \|\mu\|_{\mathcal{M}(U)} - \frac{3\varepsilon}{4}. \end{aligned}$$

This is stronger than the required lower bound. $\square$

Theorem 3.9 (Representation formula for the nonconvex functional). *Suppose that $\phi : [0, \infty) \rightarrow [0, \infty)$ is nondecreasing and concave, $\phi(0) = 0$, and*

$$L_\phi := \phi'(0+) \in (0, \infty).$$

Then, for every $\mu \in \mathcal{M}(\Omega)$,

$$(11) \quad \Phi_\phi(\mu) = \sup \left\{ \sum_{i=1}^m \phi \left(\left| \int_\Omega f_i d\mu \right| \right) : \begin{array}{l} m \geq 1, \quad f_i \in C_c(\Omega), \quad |f_i| \leq 1, \\ \text{supp}(f_i) \text{ are pairwise disjoint} \end{array} \right\}.$$

Proof. We prove the representation by first bounding every admissible finite sum from above and then constructing sums that approach $\Phi_\phi(\mu)$ from below. The hypotheses are exactly the standing hypotheses of Lemma 3.6 together with $L_\phi \in (0, \infty)$, so parts (1)–(3) of that lemma give

$$(12) \quad \phi(r+s) \leq \phi(r) + \phi(s), \quad \phi(r) \leq L_\phi r, \quad \frac{\phi(r)}{r} \longrightarrow L_\phi \quad (r \downarrow 0),$$

and part (4) gives the countable form $\phi(\sum_{j \geq 1} t_j) \leq \sum_{j \geq 1} \phi(t_j)$.

Write the atomic part of μ as

$$\mu_{\text{atom}} = \sum_{j \geq 1} c_j \delta_{\omega_j},$$

where the ω_j are distinct and $c_j \neq 0$. For an admissible family $(f_i)_{i=1}^m$, set

$$A_i := \sum_{j \geq 1} |c_j| |f_i(\omega_j)|, \quad B_i := \int_{\Omega} |f_i| d|\mu_{\text{cont}}|.$$

$$\begin{aligned} \phi \left(\left| \int_{\Omega} f_i d\mu \right| \right) &\leq \phi(A_i + B_i) && \text{by the triangle inequality} \\ &\leq \sum_{\omega_j \in \text{supp}(f_i)} \phi(|c_j|) + L_{\phi} B_i && \text{by Lemma 3.6(2), (4) and } |f_i| \leq 1. \end{aligned}$$

The disjoint supports count each atom at most once and satisfy $\sum_i |f_i| \leq 1$. Therefore

$$\begin{aligned} \sum_{i=1}^m \phi \left(\left| \int_{\Omega} f_i d\mu \right| \right) &\leq \sum_{j \geq 1} \phi(|c_j|) + L_{\phi} \int_{\Omega} \sum_{i=1}^m |f_i| d|\mu_{\text{cont}}| \\ &\leq \sum_{j \geq 1} \phi(|c_j|) + L_{\phi} \|\mu_{\text{cont}}\|_{\mathcal{M}(\Omega)} = \Phi_{\phi}(\mu). \end{aligned}$$

This proves the upper bound in (11).

For the reverse inequality, the claim is immediate when $\mu = 0$. Otherwise, fix $\varepsilon > 0$. Choose $\theta \in (0, L_{\phi})$ such that $\theta \|\mu\|_{\mathcal{M}(\Omega)} < \varepsilon/4$. By the last limit in (12), choose $\delta > 0$ such that

$$(13) \quad \phi(r) \geq (L_{\phi} - \theta)r \quad (0 \leq r \leq \delta).$$

Let

$$F := \{j \geq 1 : |c_j| > \delta/8\}, \quad \mu_{\text{rem}} := \mu - \sum_{j \in F} c_j \delta_{\omega_j}.$$

The set F is finite and every point mass of μ_{rem} has magnitude at most $\delta/8$.

Because F is finite, its points have pairwise disjoint relatively compact neighborhoods. Moreover, $\mu_{\text{rem}}(\{\omega_j\}) = 0$ for $j \in F$, so continuity of the measure from above allows these neighborhoods to be chosen with arbitrarily small total $|\mu_{\text{rem}}|$ -mass. Using the distance-function construction from Lemma 3.8, choose $g_j \in C_c(\Omega)$, $j \in F$, with pairwise disjoint supports, $|g_j| \leq 1$, and

$$g_j(\omega_j) = \text{sign}(c_j), \quad \sum_{j \in F} |\mu_{\text{rem}}|(\text{supp}(g_j)) < \tau,$$

where $\tau > 0$ can be chosen arbitrarily small. For every $j \in F$,

$$\left| \left| \int_{\Omega} g_j d\mu \right| - |c_j| \right| \leq |\mu_{\text{rem}}|(\text{supp}(g_j)).$$

By continuity of ϕ and finiteness of F , first choose τ sufficiently small and then choose the supports so that

$$(14) \quad \sum_{j \in F} \phi \left(\left| \int_{\Omega} g_j d\mu \right| \right) \geq \sum_{j \in F} \phi(|c_j|) - \frac{\varepsilon}{4}, \quad 2L_{\phi}\tau < \frac{\varepsilon}{4}.$$

Set

$$U := \Omega \setminus \bigcup_{j \in F} \text{supp}(g_j) \quad \text{and} \quad \lambda := \mu_{\text{rem}}|_U.$$

Lemma 3.8 applied on the open set U gives $h_1, \dots, h_m \in C_c(U)$ whose zero extensions belong to $C_c(\Omega)$, have pairwise disjoint supports, and satisfy

$$\left| \int_U h_i d\lambda \right| \leq \delta, \quad \sum_{i=1}^m \left| \int_U h_i d\lambda \right| \geq \|\lambda\|_{\mathcal{M}(U)} - \tau \geq \|\mu_{\text{rem}}\|_{\mathcal{M}(\Omega)} - 2\tau.$$

Together with the g_j , the zero extensions form an admissible family in (11). Since every h_i is supported in U and $\mu|_U = \lambda$, its integrals against μ and λ agree. Therefore (13) and (14) yield

$$\begin{aligned}
& \sum_{j \in F} \phi \left(\left| \int_{\Omega} g_j d\mu \right| \right) + \sum_{i=1}^m \phi \left(\left| \int_{\Omega} h_i d\mu \right| \right) \\
& \geq \sum_{j \in F} \phi(|c_j|) - \frac{\varepsilon}{4} + (L_{\phi} - \theta)(\|\mu_{\text{rem}}\|_{\mathcal{M}(\Omega)} - 2\tau) \\
& \geq \Phi_{\phi}(\mu) - \frac{\varepsilon}{4} - \theta\|\mu_{\text{rem}}\|_{\mathcal{M}(\Omega)} - 2L_{\phi}\tau \quad \text{by } \Phi_{\phi}(\mu_{\text{rem}}) \leq L_{\phi}\|\mu_{\text{rem}}\|_{\mathcal{M}(\Omega)} \\
& > \Phi_{\phi}(\mu) - \varepsilon.
\end{aligned}$$

Taking the supremum and then letting $\varepsilon \downarrow 0$ proves the reverse inequality. $\square$

Corollary 3.10. *Under the assumptions of Theorem 3.9, the functional Φ_{ϕ} is weak-* lower semicontinuous on $\mathcal{M}(\Omega) = C_0(\Omega)^*$. If ϕ is also coercive, then every sublevel of Φ_{ϕ} is bounded in total variation.*

Proof. For each fixed admissible family in (11), weak-* convergence gives convergence of all its integrals, and continuity of ϕ gives convergence of the corresponding finite sum. The representation is a supremum of weak-* continuous functions and is therefore weak-* lower semicontinuous.

For the sublevel bound, apply repeated subadditivity to the first N atoms and let $N \rightarrow \infty$. The atom magnitudes are summable, so continuity of ϕ gives

$$\phi(\|\mu_{\text{atom}}\|_{\mathcal{M}(\Omega)}) \leq \sum_{\omega \in \text{atom } \mu} \phi(|\mu(\{\omega\})|) \leq \Phi_{\phi}(\mu), \quad L_{\phi}\|\mu_{\text{cont}}\|_{\mathcal{M}(\Omega)} \leq \Phi_{\phi}(\mu).$$

If $\Phi_{\phi}(\mu) \leq C$, coercivity bounds $\|\mu_{\text{atom}}\|_{\mathcal{M}(\Omega)}$ in terms of C , while $\|\mu_{\text{cont}}\|_{\mathcal{M}(\Omega)} \leq C/L_{\phi}$. The atomic part is supported on the countable atom set, while the continuous part assigns that set zero variation. The two parts are therefore mutually singular, so their total-variation norms add to $\|\mu\|_{\mathcal{M}(\Omega)}$. Hence the sublevel is bounded in total variation. $\square$

3.3. The normalized problem. For $\alpha > 0$, define

$$J_0(\mu) := L(\mu) + \alpha\Phi_{\phi}(\mu)$$

for the unnormalized problem on the parameter domain.

Escape on the unbounded parameter domain. For positively homogeneous profiles the radial scale of ω is absorbed by the outer coefficient, and the parameter can be restricted to the unit sphere $\mathbb{S}^d$ (Section 4). But in general the domain is $\Omega = \mathbb{R}^{d+1}$, which is not compact. The regularizer Φ_{ϕ} depends on the sizes of the coefficients but not on the locations of the atoms. Along a sequence with $|\omega| \rightarrow \infty$, the fidelity need not be lower semicontinuous.

Example 3.11 (Failure of weak-* lower semicontinuity for $\tanh$). Let $\rho = \tanh$, $V \equiv 1$, fix $a \in \mathbb{R}^d$, and set $\mu_n = \delta_{(a, b_n)}$ with $b_n \rightarrow \infty$. Then $\mu_n \xrightarrow{*} 0$, yet

$$\mathcal{N}\mu_n = \tanh(a \cdot x + b_n) \rightarrow 1$$

strongly in $H^1(D)$. Hence $\liminf_{n \rightarrow \infty} L(\mu_n) = 0 < L(0)$, so the fidelity term is not weak-* lower semicontinuous. Mass escaping to infinity carries a non-vanishing value contribution unless $K(\omega)$ tends to zero as $|\omega| \rightarrow \infty$.

Note that $\Phi_{\phi}(\mu_n) = \phi(1)$ is constant along this sequence, so the unnormalized coefficient penalty does not prevent this escape. The defect is that the unnormalized kernel does not vanish at parameter infinity. The normalized atom introduced in Theorem 3.13 does vanish under the corresponding polynomial growth gap.

Example 3.12 (Nonattainment for softplus). For the profile $\rho(z) = \log(1 + e^z)$ (softplus), let $b_n \rightarrow \infty$. The measures $\mu_n = \rho(b_n)^{-1} \delta_{(0, b_n)}$ fit $V \equiv 1$ exactly for every n at penalty $\alpha \phi(1/\rho(b_n)) \rightarrow 0$, so $\inf J_0 = 0$ is not attained. Indeed, if $J_0(\mu) = 0$, then $\Phi_\phi(\mu) = 0$. Strict positivity of ϕ on $(0, \infty)$ and $L_\phi > 0$ force both the atomic and continuous parts of μ to vanish. Thus $\mu = 0$, but $L(0) = \frac{1}{2} \|1\|_{H^1(D)}^2 > 0$, a contradiction.

We need a penalty that is aware of the location of the atom to prevent the escape. The space of measure of finite moment is the right space to work with. For $p > 0$, let w_p and $\mathcal{M}_p(\Omega)$ be as in (2). For $\mu \in \mathcal{M}_p(\Omega)$, we define the normalized measure and normalized atom by

$$d\mu_p := w_p d\mu, \quad K_p(\omega) := \frac{K(\omega)}{w_p(\omega)}.$$

Theorem 3.13. *Multiplication by w_p is a linear isometric bijection*

$$\mathcal{M}_p(\Omega) \longrightarrow \mathcal{M}(\Omega), \quad \mu \longmapsto \mu_p,$$

with inverse $d\mu = w_p^{-1} d\mu_p$, and

$$(15) \quad \|\mu\|_{\mathcal{M}_p(\Omega)} = \|\mu_p\|_{\mathcal{M}(\Omega)}.$$

Moreover, μ_n converges weak-* to μ in $C_{w_p^{-1}}(\Omega)^*$ if and only if $(\mu_n)_p \xrightarrow{*} \mu_p$ in $C_0(\Omega)^*$. The multiplication preserves the atom set and the polar sign, and

$$(16) \quad (\mu_p)_{\text{atom}} = w_p \mu_{\text{atom}}, \quad (\mu_p)_{\text{cont}} = w_p \mu_{\text{cont}}.$$

If part (1) of Assumption 3.1 holds and $p > s_0$, then $K_p \in C_0(\Omega; L^2(D))$ and

$$(17) \quad \mathcal{N}\mu = \int_{\Omega} K d\mu = \int_{\Omega} K_p d\mu_p.$$

Finally,

$$(18) \quad \Phi_\phi(\mu_p) = L_\phi \int_{\Omega} w_p d|\mu_{\text{cont}}| + \sum_{\omega \in \text{atom } \mu} \phi(w_p(\omega) |\mu(\{\omega\})|).$$

Proof. The isometric identification with $\mathcal{M}(\Omega)$, including its inverse, the norm identity, and the weak-* equivalence, is the polynomial-weight specialization of [3, Section 5.1]; in particular, this gives (15). Since multiplication by the strictly positive function w_p preserves null sets, the atom set, the polar sign, and nonatomicity, (16) follows immediately.

It remains to treat the network map. The continuity of K_p follows from that of K and w_p . For $r = |\omega| \geq 1$, the growth estimate (3) gives

$$\begin{aligned} \|K_p(\omega)\|_{L^2(D)} &\leq C_D \frac{(1+r)^{s_0}}{1+r^p} \\ &\leq 2^{s_0} C_D r^{s_0-p} \longrightarrow 0, \quad r \rightarrow \infty, \end{aligned}$$

because $p > s_0$. Thus $K_p \in C_0(\Omega; L^2(D))$. The change-of-measure identity for the Bochner integral now gives (17). Finally, (16) and $|w_p \mu_{\text{cont}}| = w_p |\mu_{\text{cont}}|$ give (18) directly from (7). $\square$

For $\alpha > 0$, define the adopted objective

$$J(\mu) := L(\mu) + \alpha \Phi_\phi(\mu_p), \quad \mu \in \mathcal{M}_p(\Omega).$$

For a finite atomic measure $\mu_{\omega, c} = \sum_{n=1}^N c_n \delta_{\omega_n}$ with distinct support points, this becomes

$$J(\mu_{\omega, c}) = L(\mu_{\omega, c}) + \alpha \sum_{n=1}^N \phi(w_p(\omega_n) |c_n|).$$

Throughout, finite-network representations are understood after repeated locations are merged and zero coefficients are discarded; their width is therefore $\# \text{ atom } \mu$.

Definition 3.14 (Local minimizer). A measure $\bar{\mu} \in \mathcal{M}_p(\Omega)$ is a local minimizer of J if $J(\bar{\mu}) < \infty$ and there is $\varepsilon > 0$ such that

$$J(\bar{\mu}) \leq J(\mu) \quad \text{for all} \quad \mu \in \mathcal{M}_p(\Omega), \quad \|\mu - \bar{\mu}\|_{\mathcal{M}_p(\Omega)} < \varepsilon.$$

By Theorem 3.13, this is exactly total-variation local minimality in the normalized coordinate μ_p .

3.4. Existence of a global minimizer. We apply the direct method in the normalized coordinate. Weak-* compactness controls μ_p , Corollary 3.10 supplies the penalty lower semicontinuity and sublevel bound, and Lemma 3.15 gives the strong convergence needed for the value part of the fidelity.

Lemma 3.15. Assume (3), let $p > s_0$, and suppose that

$$(\mu_n)_p \xrightarrow{*} \mu_p \quad \text{in } \mathcal{M}(\Omega).$$

Then $\mathcal{N}\mu_n \rightarrow \mathcal{N}\mu$ strongly in $L^2(D)$.

Proof. We use a compactly supported cutoff to pass to the weak-* limit locally and then control the tails uniformly. By Theorem 3.13,

$$\mathcal{N}\mu_n = \int_{\Omega} K_p d(\mu_n)_p, \quad \mathcal{N}\mu = \int_{\Omega} K_p d\mu_p.$$

Set

$$C := \sup_n \|(\mu_n)_p\|_{\mathcal{M}(\Omega)} < \infty,$$

where finiteness follows from weak-* convergence and the uniform boundedness principle. Weak-* lower semicontinuity of total variation gives $\|\mu_p\|_{\mathcal{M}(\Omega)} \leq C$.

For $R \geq 1$, choose $\chi_R \in C_c(\Omega)$ with $0 \leq \chi_R \leq 1$, $\chi_R = 1$ on B_R , and $\text{supp } \chi_R \subset B_{R+1}$. For fixed R and $x \in D$, the function

$$\omega \mapsto \chi_R(\omega)[K_p(\omega)](x)$$

belongs to $C_c(\Omega)$. Hence weak-* convergence gives

$$\int_{\Omega} \chi_R(\omega)[K_p(\omega)](x) d(\mu_n)_p(\omega) \longrightarrow \int_{\Omega} \chi_R(\omega)[K_p(\omega)](x) d\mu_p(\omega)$$

for every $x \in D$.

Since D is bounded and $\text{supp } \chi_R$ is compact,

$$M_R := \sup_{\omega \in \text{supp } \chi_R, x \in D} |[K_p(\omega)](x)| < \infty.$$

The absolute values of both truncated integrals are bounded by CM_R . The integrand $(\omega, x) \mapsto \chi_R(\omega)[K_p(\omega)](x)$ is jointly Borel measurable and bounded by M_R ; Fubini's theorem therefore identifies the pointwise integrals above, almost everywhere in x , with representatives of the $L^2(D)$ -valued Bochner integrals below. The constant function $2CM_R$ belongs to $L^2(D)$ because D is bounded. Dominated convergence therefore yields

$$\int_{\Omega} \chi_R K_p d(\mu_n)_p \longrightarrow \int_{\Omega} \chi_R K_p d\mu_p \quad \text{strongly in } L^2(D).$$

It remains to remove the cutoff. If $r := |\omega| > R \geq 1$, then

$$\begin{aligned} \|K_p(\omega)\|_{L^2(D)} &\leq C_D \frac{(1+r)^{s_0}}{1+r^p} && \text{by (3)} \\ &\leq 2^{s_0} C_D r^{s_0-p} \leq 2^{s_0} C_D R^{s_0-p}, && \text{since } p > s_0 \text{ and } r > R. \end{aligned}$$

Consequently,

$$\left\| \int_{\Omega} (1 - \chi_R) K_p d(\mu_n)_p \right\|_{L^2(D)} \leq 2^{s_0} C_D C R^{s_0-p},$$

$$\left\| \int_{\Omega} (1 - \chi_R) K_p d\mu_p \right\|_{L^2(D)} \leq 2^{s_0} C_D C R^{s_0-p}.$$

For fixed R , let $n \rightarrow \infty$ in the truncated integrals. Then let $R \rightarrow \infty$; the two tail bounds vanish because $s_0 - p < 0$. This proves the claim. $\square$

Theorem 3.16 (Existence on the unbounded parameter domain). *Let $V \in H^1(D)$, let ϕ satisfy Assumptions 3.2 and 3.3, and let $\rho \in C(\mathbb{R})$ satisfy Assumption 3.1(1). If $p > s_0$ and $\alpha > 0$, then*

$$\inf_{\mu \in \mathcal{M}_p(\Omega)} J(\mu)$$

admits at least one global minimizer.

Proof. We use the direct method in the normalized measure coordinate. Only the stated continuity and value-growth bound for ρ are needed. Since $V \in H^1(D)$,

$$J(0) = \frac{1}{2} \|V\|_{H^1(D)}^2 =: C < \infty.$$

Choose a minimizing sequence $(\mu_n) \subset \mathcal{M}_p(\Omega)$ with $J(\mu_n) \leq C + 1$. The fidelity and penalty are nonnegative, so

$$\Phi_{\phi}((\mu_n)_p) \leq \frac{C + 1}{\alpha}.$$

The total-variation sublevel bound in Corollary 3.10 shows that $((\mu_n)_p)$ is bounded in $\mathcal{M}(\Omega)$. Weak-* compactness and separability of $C_0(\Omega)$ give, after passing to a subsequence,

$$(\mu_n)_p \xrightarrow{*} \bar{\mu}_p.$$

Theorem 3.13 defines a unique $\bar{\mu} \in \mathcal{M}_p(\Omega)$ with normalized coordinate $\bar{\mu}_p$. Lemma 3.15 gives

$$(19) \quad \mathcal{N}\mu_n \longrightarrow \mathcal{N}\bar{\mu} \quad \text{strongly in } L^2(D).$$

The fidelity bound gives

$$\sup_n \|\nabla(\mathcal{N}\mu_n)\|_{L^2(D)} \leq \|\nabla V\|_{L^2(D)} + \sqrt{2(C + 1)}.$$

After another subsequence, $\nabla(\mathcal{N}\mu_n) \rightharpoonup g$ weakly in $L^2(D)^d$. For every $\psi \in C_c^\infty(D)^d$,

$$\int_D g \cdot \psi dx = \lim_{n \rightarrow \infty} \int_D \nabla(\mathcal{N}\mu_n) \cdot \psi dx = - \lim_{n \rightarrow \infty} \int_D \mathcal{N}\mu_n \operatorname{div} \psi dx = - \int_D \mathcal{N}\bar{\mu} \operatorname{div} \psi dx,$$

the last step by (19). Hence $g = \nabla(\mathcal{N}\bar{\mu})$ distributionally and $\mathcal{N}\bar{\mu} \in H^1(D)$.

The value part of the fidelity is continuous by (19), and the gradient part is weakly lower semicontinuous. Hence

$$L(\bar{\mu}) \leq \liminf_{n \rightarrow \infty} L(\mu_n).$$

Corollary 3.10 gives

$$\Phi_{\phi}(\bar{\mu}_p) \leq \liminf_{n \rightarrow \infty} \Phi_{\phi}((\mu_n)_p).$$

Combining the two inequalities proves $J(\bar{\mu}) \leq \liminf_n J(\mu_n) = \inf J$. $\square$

3.5. Optimality conditions and finite support. Let $\bar{\mu}$ be a local minimizer. For a measure μ and an admissible direction μ' , the Gâteaux derivative is

$$DL(\mu)[\mu'] := \lim_{t \rightarrow 0} \frac{L(\mu + t\mu') - L(\mu)}{t} = \langle r_{\mu}, \mathcal{N}\mu' \rangle_{H^1(D)}.$$

The corresponding derivative profile is

$$P_{\mu}(\omega) := DL(\mu)[\delta_{\omega}] = \langle r_{\mu}, K(\omega) \rangle_{H^1(D)}.$$

Define its normalized form and the uniform normalized feature bound by

$$P_{p,\mu}(\omega) := \frac{P_{\mu}(\omega)}{w_p(\omega)} = \langle r_{\mu}, K_p(\omega) \rangle_{H^1(D)}, \quad B_p := \sup_{\omega \in \Omega} \|K_p(\omega)\|_{H^1(D)}.$$

At the local minimizer write $\bar{P} := P_{\bar{\mu}}$ and $\bar{P}_p := P_{p, \bar{\mu}}$. Thus $\bar{P}$ is the function representing the Gâteaux derivative. For differentiable ρ ,

$$\bar{P}(a, b) = \int_D r_{\bar{\mu}}(x) \rho(a \cdot x + b) dx + \int_D \nabla r_{\bar{\mu}}(x) \cdot a \rho'(a \cdot x + b) dx.$$

The second term carries the factor a , so on the unbounded domain $\bar{P}$ is defined pointwise but need not be bounded.

By Cauchy–Schwarz, (4) gives

$$|\bar{P}(\omega)| \leq \|r_{\bar{\mu}}\|_{H^1(D)} \|K(\omega)\|_{H^1(D)} \leq C_P (1 + |\omega|)^{s_1}, \quad C_P := C_K \|r_{\bar{\mu}}\|_{H^1(D)}.$$

The constant C_P depends on the fixed minimizer through its residual, but not on ω . If $p > s_1$, then

$$K_p \in C_0(\Omega; H^1(D)), \quad \bar{P}_p \in C_0(\Omega), \quad \bar{P} \in C_{w_p^{-1}}(\Omega).$$

Consequently, for every $\mu' \in \mathcal{M}_p(\Omega)$,

$$DL(\bar{\mu})[\mu'] = \int_{\Omega} \bar{P} d\mu' = \int_{\Omega} \bar{P}_p d\mu'_p.$$

The pointwise definition of $\bar{P}$ as a Gâteaux derivative does not require $\bar{P} \in C_0(\Omega)$.

Theorem 3.17 (The optimality condition). *Let $\phi : [0, \infty) \rightarrow [0, \infty)$ be concave and nondecreasing, with $\phi(0) = 0$, $\phi \in C^1((0, \infty))$, and $0 < L_\phi := \phi'(0+) < \infty$. Let $\bar{\mu} \in \mathcal{M}_p(\Omega)$ be a local minimizer of J with $J(\bar{\mu}) < \infty$, and suppose that Assumption 3.1 holds and $p > s_1$. Then*

$$(20) \quad |\bar{P}_p(\omega)| \leq \alpha L_\phi \quad \text{for every } \omega \in \Omega,$$

$$(21) \quad \bar{P}_p(\omega) = -\alpha \phi'(w_p(\omega) |\bar{\mu}(\{\omega\})|) \operatorname{sign}(\bar{\mu}(\{\omega\})) \quad \text{for } \omega \in \operatorname{atom} \bar{\mu}.$$

Moreover,

$$(22) \quad \bar{P}_p = -\alpha L_\phi \operatorname{sign}(\bar{\mu}_{\text{cont}}) \quad |(\bar{\mu}_p)_{\text{cont}}| \text{-almost everywhere.}$$

Proof. We use Theorem 3.13 to work with $\bar{\mu}_p$ in the total-variation norm. The normalized fidelity derivative in the direction $\xi \in \mathcal{M}(\Omega)$ is

$$\left\langle r_{\bar{\mu}}, \int_{\Omega} K_p d\xi \right\rangle_{H^1(D)} = \int_{\Omega} \bar{P}_p d\xi.$$

Fix $\omega \in \operatorname{atom} \bar{\mu}$ and put $z := \bar{\mu}_p(\{\omega\}) = w_p(\omega) \bar{\mu}(\{\omega\}) \neq 0$. For $s \in \{-1, 1\}$ and sufficiently small $t > 0$, the sign of $z + ts$ equals the sign of z . Dividing the local-minimality inequality by t and letting $t \downarrow 0$ gives

$$0 \leq s [\bar{P}_p(\omega) + \alpha \phi'(|z|) \operatorname{sign}(z)].$$

Using both choices of s makes the bracket vanish, and hence

$$\bar{P}_p(\omega) = -\alpha \phi'(|z|) \operatorname{sign}(z).$$

Because $w_p > 0$, $\operatorname{sign}(z) = \operatorname{sign}(\bar{\mu}(\{\omega\}))$, so this is (21). Concavity and monotonicity give $0 \leq \phi'(|z|) \leq L_\phi$, and hence $|\bar{P}_p(\omega)| \leq \alpha L_\phi$ at every atom.

If $\omega \notin \operatorname{atom} \bar{\mu}$, insertion of $ts\delta_\omega$ into the normalized measure changes the penalty by $\phi(t)$. Dividing the local minimality inequality by $t > 0$ and letting $t \downarrow 0$ yields

$$0 \leq s \bar{P}_p(\omega) + \alpha L_\phi.$$

The two choices of s prove (20) away from the atoms, and the preceding atom estimate completes its proof.

It remains to identify the continuous part. Set $\xi := (\bar{\mu}_p)_{\text{cont}}$. Since $K_p \in C_0(\Omega; H^1(D))$, the directions $\pm\xi$ are admissible. For $s \in \{-1, 1\}$ and $0 < t < 1$, the perturbation $\bar{\mu}_p + ts\xi$ multiplies the continuous part by $1 + ts > 0$. Its one-sided directional derivative therefore gives

$$0 \leq s \left[\int_{\Omega} \bar{P}_p d\xi + \alpha L_{\phi} \|\xi\|_{\mathcal{M}(\Omega)} \right].$$

Using both signs shows that the bracket vanishes. On the other hand, (20) makes

$$\bar{P}_p \text{sign}(\xi) + \alpha L_{\phi} \geq 0$$

$|\xi|$ -almost everywhere. Its integral is zero, so the integrand vanishes $|\xi|$ -almost everywhere. The sign preservation in Theorem 3.13 gives (22). $\square$

Remark 3.18. Because $\bar{P}_p \in C_0(\Omega)$ and $\alpha L_{\phi} > 0$, every closed positive superlevel set of $|\bar{P}_p|$ is compact. Thus any violation of (20) is confined to a compact set, even when the unnormalized function $\bar{P}$ does not decay at parameter infinity. Lemma 3.19 uses a lower superlevel to localize the entire support.

Lemma 3.19. *Let ϕ satisfy Assumptions 3.2 and 3.3, let $p > s_1$, and suppose that Assumption 3.1 holds. Suppose that $\bar{\mu} \in \mathcal{M}_p(\Omega)$ is an $\mathcal{M}_p(\Omega)$ -local minimizer of J such that $0 < J(\bar{\mu}) < \infty$. Define*

$$T(\bar{\mu}) := \phi^{-1} \left(\frac{J(\bar{\mu})}{\alpha} \right)$$

and

$$\mathcal{K}(\bar{\mu}) := \{\omega \in \Omega : |\bar{P}_p(\omega)| \geq \alpha \phi'(T(\bar{\mu}))\}.$$

Then $\mathcal{K}(\bar{\mu})$ is compact and

$$(23) \quad \text{supp } \bar{\mu}_{\text{atom}} \cup \text{supp } \bar{\mu}_{\text{cont}} \subset \mathcal{K}(\bar{\mu}).$$

Moreover, every atom $\omega \in \text{atom } \bar{\mu}$ satisfies

$$w_p(\omega) |\bar{\mu}(\{\omega\})| \leq T(\bar{\mu}).$$

Proof. We use Theorem 3.17 and the fact that $\bar{P}_p$ vanishes at infinity to prove the localization. Concavity and monotonicity make ϕ' positive and nonincreasing, as observed after Assumption 3.3. Since $\bar{P}_p \in C_0(\Omega)$, its closed superlevel set at the strictly positive level $\alpha \phi'(T(\bar{\mu}))$ is compact. This proves the first assertion.

Let $\omega \in \text{atom } \bar{\mu}$ and put

$$t_{\omega} := |\bar{\mu}_p(\{\omega\})| = w_p(\omega) |\bar{\mu}(\{\omega\})|.$$

The contribution of this atom to the objective gives

$$\begin{aligned} \alpha \phi(t_{\omega}) &\leq \alpha \Phi_{\phi}(\bar{\mu}_p) && \text{by nonnegativity of the other terms} \\ &\leq J(\bar{\mu}) = \alpha \phi(T(\bar{\mu})). \end{aligned}$$

Strict monotonicity of ϕ gives $t_{\omega} \leq T(\bar{\mu})$. The atom condition in Theorem 3.17 and monotonicity of ϕ' now give

$$|\bar{P}_p(\omega)| = \alpha \phi'(t_{\omega}) \geq \alpha \phi'(T(\bar{\mu})).$$

Thus every atom lies in $\mathcal{K}(\bar{\mu})$.

On the continuous part, Theorem 3.17 gives

$$|\bar{P}_p(\omega)| = \alpha L_{\phi} \geq \alpha \phi'(T(\bar{\mu})) \quad |(\bar{\mu}_p)_{\text{cont}}| \text{-almost everywhere.}$$

Hence $(\bar{\mu}_p)_{\text{cont}}$ is concentrated on the closed set $\mathcal{K}(\bar{\mu})$. Theorem 3.13 preserves the atomic and continuous decompositions. Since w_p is strictly positive, each component of $\bar{\mu}_p = w_p \bar{\mu}$ has the same null sets, and hence the same support, as the corresponding component of $\bar{\mu}$. Taking closures for the atomic part and supports for the continuous part proves (23). $\square$

Theorem 3.20 (Finite support and separation of local minimizers). *Let ϕ satisfy Assumptions 3.2 and 3.3, suppose that $\phi \in C^2((0, \infty))$ satisfies Assumption 3.4(2), let $p > s_1$, and suppose that Assumption 3.1 holds. Then every $\mathcal{M}_p(\Omega)$ -local minimizer $\bar{\mu} \in \mathcal{M}_p(\Omega)$ of J with $J(\bar{\mu}) < \infty$ is finite atomic. If $J(\bar{\mu}) > 0$, set*

$$T := T(\bar{\mu}), \quad \mathcal{K} := \mathcal{K}(\bar{\mu}).$$

Then $\bar{\mu}_{\text{cont}} = 0$, every atom belongs to the compact set $\mathcal{K}$, and any two distinct atoms ω_i and ω_j satisfy

$$(24) \quad \|K_p(\omega_i) - K_p(\omega_j)\|_{H^1(D)}^2 \geq 2\alpha\gamma_T.$$

Consequently,

$$(25) \quad \#\text{atom}(\bar{\mu}) \leq N_{\text{cov}}\left(K_p(\mathcal{K}), \frac{1}{2}\sqrt{\alpha\gamma_T}\right) < \infty,$$

where the covering number $N_{\text{cov}}(A, r)$ is the least number of balls of radius r needed to cover A .

Proof. We use the isometry of Theorem 3.13, the first-order identities of Theorem 3.17, and local coefficient variations to prove the assertions. Concavity, coercivity, and $L_\phi > 0$ imply that $\phi(t) > 0$ for $t > 0$.

The zero objective case is immediate:

$$J(\bar{\mu}) = 0 \implies \Phi_\phi(\bar{\mu}_p) = 0 \implies \bar{\mu}_p = 0 \implies \bar{\mu} = 0,$$

where the middle implication follows from $L_\phi > 0$ and $\phi(t) > 0$ for $t > 0$. We may therefore suppose $J(\bar{\mu}) > 0$. Lemma 3.19 gives the compact set $\mathcal{K}$ and already localizes the atomic and continuous supports. Every normalized atom magnitude

$$t_i := |\bar{\mu}_p(\{\omega_i\})|$$

satisfies $t_i \leq T$ by Lemma 3.19. The curvature assumption gives $\phi''(t) \leq -\gamma_T$ for $0 < t \leq T$. Integrating this inequality from the origin, using $\phi'(0+) = L_\phi$, and then integrating once more gives

$$(26) \quad L_\phi t - \phi(t) \geq \frac{\gamma_T}{2} t^2 \quad (0 \leq t \leq T).$$

We first separate distinct atoms. For two such atoms, Theorem 3.13 defines a unique measure $\mu_s \in \mathcal{M}_p(\Omega)$ by

$$(\mu_s)_p := \bar{\mu}_p + s\delta_{\omega_i} - s\delta_{\omega_j}.$$

For sufficiently small $|s|$, the signs of both atom coefficients remain fixed, and

$$\|\mu_s - \bar{\mu}\|_{\mathcal{M}_p(\Omega)} = 2|s|, \quad \text{by Theorem 3.13,}$$

$$\mathcal{N}\mu_s = \mathcal{N}\bar{\mu} + s[K_p(\omega_i) - K_p(\omega_j)], \quad \text{by the normalized representation of } \mathcal{N}.$$

Thus $s \mapsto J(\mu_s)$ is twice differentiable near zero and has a local minimum there. Its second derivative satisfies

$$\begin{aligned} 0 &\leq \left. \frac{d^2}{ds^2} J(\mu_s) \right|_{s=0} \\ &= \|K_p(\omega_i) - K_p(\omega_j)\|_{H^1(D)}^2 + \alpha\phi''(t_i) + \alpha\phi''(t_j) && \text{by the quadratic fidelity} \\ &\leq \|K_p(\omega_i) - K_p(\omega_j)\|_{H^1(D)}^2 - 2\alpha\gamma_T, && \text{since } t_i, t_j \leq T. \end{aligned}$$

This proves (24).

We next rule out the continuous part by concentrating a local piece of one sign. Suppose $(\bar{\mu}_p)_{\text{cont}} \neq 0$ and write

$$d(\bar{\mu}_p)_{\text{cont}} = \sigma d\lambda, \quad \lambda := |(\bar{\mu}_p)_{\text{cont}}|,$$

where σ takes the values -1 and 1 λ -almost everywhere. Choose $\sigma_0 \in \{-1, 1\}$ such that the restriction of λ to $\{\sigma = \sigma_0\}$ is nonzero, and denote that restriction again by λ . Theorem 3.17

gives $\bar{P}_p = -\alpha L_\phi \sigma_0$ on a set of full λ -measure. The support of a Radon measure has full measure, while the countable set $\text{atom } \bar{\mu}$ has zero λ -measure because λ is nonatomic. The intersection of these three full λ -measure sets is nonempty, so choose

$$\bar{\omega} \in \text{supp } \lambda \setminus \text{atom } \bar{\mu} \quad \text{such that} \quad \bar{P}_p(\bar{\omega}) = -\alpha L_\phi \sigma_0.$$

For $\delta > 0$, put

$$r_\delta := \lambda(B_\delta(\bar{\omega})), \quad \varepsilon_\delta := \sup_{\omega \in B_\delta(\bar{\omega})} \|K_p(\omega) - K_p(\bar{\omega})\|_{H^1(D)}.$$

Because $\bar{\omega} \in \text{supp } \lambda$, one has $r_\delta > 0$. Moreover, $r_\delta \rightarrow \lambda(\{\bar{\omega}\}) = 0$ by continuity from above and the nonatomicity of λ , while $\varepsilon_\delta \rightarrow 0$ by continuity of K_p at $\bar{\omega}$. Define μ_δ through its normalized coordinate by

$$(\mu_\delta)_p := \bar{\mu}_p - \sigma_0 \lambda|_{B_\delta(\bar{\omega})} + \sigma_0 r_\delta \delta_{\bar{\omega}}.$$

The target point is not an atom, and Theorem 3.13 gives

$$\|\mu_\delta - \bar{\mu}\|_{\mathcal{M}_p(\Omega)} = 2r_\delta \rightarrow 0.$$

If $e_\delta := \mathcal{N}(\mu_\delta - \bar{\mu})$, then

$$\|e_\delta\|_{H^1(D)} \leq \varepsilon_\delta r_\delta, \quad \text{by the Bochner-integral triangle inequality,}$$

$$\langle r_\delta, e_\delta \rangle_{H^1(D)} = \sigma_0 r_\delta \bar{P}_p(\bar{\omega}) - \sigma_0 \int_{B_\delta(\bar{\omega})} \bar{P}_p d\lambda = 0, \quad \text{by Theorem 3.17.}$$

The exact quadratic expansion of the fidelity and the definition of Φ_ϕ therefore give, for all sufficiently small δ such that $r_\delta \leq T$,

$$\begin{aligned} J(\mu_\delta) - J(\bar{\mu}) &= \frac{1}{2} \|e_\delta\|_{H^1(D)}^2 + \alpha [\phi(r_\delta) - L_\phi r_\delta] \\ &\leq \frac{1}{2} (\varepsilon_\delta^2 - \alpha \gamma_T) r_\delta^2 && \text{by (26)} \\ &< 0, && \text{for sufficiently small } \delta. \end{aligned}$$

This contradicts local minimality. Hence $\bar{\mu}_{\text{cont}} = 0$.

It remains to count the atoms. The set $K_p(\mathcal{K})$ is compact because $\mathcal{K}$ is compact and K_p is continuous. Cover it by balls of radius $\frac{1}{2}\sqrt{\alpha\gamma_T}$. Each such ball has diameter $\sqrt{\alpha\gamma_T}$, which is strictly smaller than the separation $\sqrt{2\alpha\gamma_T}$ in (24). Hence each ball contains the image of at most one atom. A compact set has a finite cover at every positive radius, and (25) follows. $\square$

Corollary 3.21. *Assume the hypotheses of Theorems 3.16 and 3.20. Then J admits a global minimizer, every global minimizer is finite atomic, and*

$$(27) \quad \min_{\mu \in \mathcal{M}_p(\Omega)} J(\mu) = \min_{\substack{N \geq 1, \omega_1, \dots, \omega_N \in \Omega \\ c_1, \dots, c_N \in \mathbb{R}}} \left\{ \frac{1}{2} \left\| \sum_{n=1}^N c_n K(\omega_n) - V \right\|_{H^1(D)}^2 + \alpha \sum_{n=1}^N \phi(w_p(\omega_n) | c_n |) \right\}.$$

The minimum on the right is attained. If μ_ is any global minimizer with $J(\mu_*) > 0$, then it satisfies the data-dependent bound*

$$(28) \quad \# \text{atom}(\mu_*) \leq N_{\text{cov}} \left(K_p(\mathcal{K}(\mu_*)), \frac{1}{2} \sqrt{\alpha\gamma_{T(\mu_*)}} \right).$$

If $V \neq 0$, define

$$T_0 := \phi^{-1} \left(\frac{\|V\|_{H^1(D)}^2}{2\alpha} \right)$$

and

$$(29) \quad \mathcal{K}_0 := \{ \omega \in \Omega : \|V\|_{H^1(D)} \|K_p(\omega)\|_{H^1(D)} \geq \alpha \phi'(T_0) \}.$$

Then $\mathcal{K}_0$ is compact, every atom of every global minimizer lies in $\mathcal{K}_0$, and

$$(30) \quad \# \text{atom}(\mu_*) \leq N_{\text{cov}} \left(K_p(\mathcal{K}_0), \frac{1}{2} \sqrt{\alpha \gamma T_0} \right) < \infty.$$

If $V = 0$, then the zero measure is the unique global minimizer.

Proof. We combine the direct method in Theorem 3.16 with the local result in Theorem 3.20. Theorem 3.16 gives a global minimizer. Every global minimizer is $\mathcal{M}_p(\Omega)$ -local, so Theorem 3.20 makes it finite atomic; when its objective is positive, Theorem 3.20 also gives (28). Its distinct atom representation has objective equal to the expression on the right of (27), so the right-hand infimum is no larger than the measure minimum.

Conversely, consider an arbitrary finite network, possibly with repeated locations. For each distinct location η_j , merge its coefficients into $d_j := \sum_{n: \omega_n = \eta_j} c_n$. The represented function, and hence the fidelity, is unchanged. Monotonicity and repeated subadditivity of ϕ give

$$\phi(w_p(\eta_j)|d_j|) \leq \phi \left(w_p(\eta_j) \sum_{n: \omega_n = \eta_j} |c_n| \right) \leq \sum_{n: \omega_n = \eta_j} \phi(w_p(\eta_j)|c_n|).$$

After summing over the distinct locations, the measure objective is no larger than the displayed finite-network objective. Thus the measure minimum is no larger than the right-hand infimum in (27). The two infima are equal, and the finite atom representation of a global minimizer attains the right-hand side, so both infima are minima.

Suppose $V \neq 0$. Then a global minimizer μ_* has positive objective, because Theorem 3.20 would otherwise give $\mu_* = 0$, whose fidelity is positive. By comparison with the zero measure,

$$\begin{aligned} J(\mu_*) &\leq J(0) = \frac{1}{2} \|V\|_{H^1(D)}^2 = \alpha \phi(T_0), \\ \|r_{\mu_*}\|_{H^1(D)} &\leq \|V\|_{H^1(D)}, \end{aligned} \quad \text{since } L(\mu_*) \leq J(\mu_*).$$

Hence

$$(31) \quad T(\mu_*) \leq T_0, \quad \gamma_{T(\mu_*)} \geq \gamma_{T_0}.$$

For an atom ω_i of normalized magnitude $t_i = |\mu_{*,p}(\{\omega_i\})|$, Lemma 3.19 gives $t_i \leq T(\mu_*)$. Theorem 3.17 then gives

$$\begin{aligned} \alpha \phi'(T_0) &\leq \alpha \phi'(t_i) = |\langle r_{\mu_*}, K_p(\omega_i) \rangle_{H^1(D)}| && \text{since } t_i \leq T(\mu_*) \leq T_0 \\ &\leq \|r_{\mu_*}\|_{H^1(D)} \|K_p(\omega_i)\|_{H^1(D)} && \text{by Cauchy-Schwarz} \\ &\leq \|V\|_{H^1(D)} \|K_p(\omega_i)\|_{H^1(D)}. \end{aligned}$$

Thus every atom belongs to $\mathcal{K}_0$. The level in (29) is strictly positive, while $K_p \in C_0(\Omega; H^1(D))$, so $\mathcal{K}_0$ is compact. Combining (31) with (24) gives pairwise separation at least $\sqrt{2\alpha\gamma T_0}$. The covering argument from Theorem 3.20 on $K_p(\mathcal{K}_0)$ proves (30).

If $V = 0$, then $J(0) = 0$. Nonnegativity makes zero the minimum, and the zero-objective conclusion of Theorem 3.20 shows that no other global minimizer exists. $\square$

The hypotheses used so far are jointly satisfiable, and the following family makes the constants explicit.

Example 3.22 (separation fails for convex penalty). For $\gamma \geq 0$, define

$$(32) \quad \phi_\gamma(z) := \begin{cases} \frac{z}{2} + \frac{1}{4\gamma} \log(1 + 2\gamma z), & \gamma > 0, \\ z, & \gamma = 0. \end{cases}$$

Then $\phi_\gamma(0) = 0$ and, for $\gamma > 0$,

$$\phi'_\gamma(z) = \frac{1}{2} + \frac{1}{2(1 + 2\gamma z)} \in \left(\frac{1}{2}, 1\right], \quad \phi''_\gamma(t) = -\frac{\gamma}{(1 + 2\gamma t)^2}.$$

Hence $L_{\phi_\gamma} = \phi'_\gamma(0) = 1 \in (0, \infty)$, which is Assumption 3.2; the derivative is positive and nonincreasing and $\phi_\gamma(z) \geq z/2 \rightarrow \infty$, which is Assumption 3.3. For Assumption 3.4, the second derivative is continuous and negative on $(0, \infty)$, part (1) holds with $\gamma_1 = \gamma$ and any $\hat{z}_1 > 0$, and part (2) holds with

$$(33) \quad \gamma_A = \frac{\gamma}{(1 + 2\gamma A)^2} > 0.$$

Thus ϕ_γ with $\gamma > 0$ satisfies Assumptions 3.2–3.4, and (33) turns the atom separation $\sqrt{2\alpha\gamma T_0}$ and the neuron bound (30) into computable quantities. This is the penalty used by Algorithm 1.

The limit $\gamma \rightarrow 0$ is the convex endpoint. Assumption 3.4 fails at $\gamma = 0$, so Theorem 3.20 and Corollary 3.21 do not apply; the weaker conclusions of Theorems 3.16, 3.17, and 5.1 remain available. The degeneration is quantitative rather than abrupt: by (33), $0 < \gamma_{T_0} \leq \gamma \rightarrow 0$ as $\gamma \rightarrow 0$. Thus the guaranteed separation scale tends to zero; the corresponding covering-number bound may deteriorate or become uninformative.

For fixed, distinct locations $\omega = (\omega_n)_{n=1}^N \in \Omega^N$, define the reduced coefficient objective

$$(34) \quad J_\omega(c) := L \left(\sum_{n=1}^N c_n \delta_{\omega_n} \right) + \alpha \sum_{n=1}^N \phi(w_p(\omega_n)|c_n|), \quad c \in \mathbb{R}^N.$$

Theorem 3.23 (Sufficient condition for local optimality). *Let $\phi : [0, \infty) \rightarrow [0, \infty)$ satisfy $\phi(0) = 0$ and $\phi \in C^1((0, \infty))$, with $0 < L_\phi := \phi'(0+) < \infty$. Suppose there are $\gamma_1 \geq 0$ and $\hat{z}_1 > 0$ such that*

$$(35) \quad \phi(z) \geq L_\phi z - \frac{\gamma_1}{2} z^2 \quad (0 \leq z \leq \hat{z}_1).$$

Let $p > s_1$, let $V \in H^1(D)$, and suppose that Assumption 3.1 holds. Let

$$\bar{\omega} := (\bar{\omega}_n)_{n=1}^N, \quad \bar{c} := (\bar{c}_n)_{n=1}^N \in (\mathbb{R} \setminus \{0\})^N, \quad \bar{\mu} := \sum_{n=1}^N \bar{c}_n \delta_{\bar{\omega}_n},$$

where the points $\bar{\omega}_n$ are distinct and every $\bar{c}_n$ is nonzero. Suppose also that

$$(36) \quad |\phi'(w_p(\bar{\omega}_n)|\bar{c}_n|)| < L_\phi \quad (n = 1, \dots, N).$$

Define

$$\bar{P}_p(\omega) := \langle \mathcal{N}\bar{\mu} - V, K_p(\omega) \rangle_{H^1(D)}.$$

Assume that

- (1) $\bar{c} \in (\mathbb{R} \setminus \{0\})^N$ is a local minimizer of $J_{\bar{\omega}}$;
- (2) the strict inequality

$$|\bar{P}_p(\omega)| < \alpha L_\phi$$

holds for every $\omega \in \Omega \setminus \{\bar{\omega}_1, \dots, \bar{\omega}_N\}$.

Then $\bar{\mu}$ is an $\mathcal{M}_p(\Omega)$ -local minimizer of J .

Proof. We use the normalized coordinate to reduce the claim to total-variation estimates. Set

$$\bar{z} := (\bar{z}_n)_{n=1}^N, \quad \bar{z}_n := w_p(\bar{\omega}_n)\bar{c}_n.$$

For $z = (z_n)_{n=1}^N \in \mathbb{R}^N$, define

$$\hat{J}(z) := \frac{1}{2} \left\| \sum_{n=1}^N z_n K_p(\bar{\omega}_n) - V \right\|_{H^1(D)}^2 + \alpha \sum_{n=1}^N \phi(|z_n|).$$

The invertible diagonal change $z_n = w_p(\bar{\omega}_n)c_n$ shows that $\bar{z}$ is a local minimizer of $\hat{J}$.

We first obtain a uniform strict gap. Stationarity of $J_{\bar{\omega}}$ at $\bar{c}$ gives

$$\bar{P}_p(\bar{\omega}_n) = -\alpha \phi'(|\bar{z}_n|) \text{sign}(\bar{z}_n).$$

The active-slope gap (36) and condition (2) give

$$|\bar{P}_p(\omega)| < \alpha L_\phi \quad (\omega \in \Omega).$$

Since $\bar{P}_p \in C_0(\Omega)$, the positive continuous function $\alpha L_\phi - |\bar{P}_p|$ tends to αL_ϕ at infinity. Hence there is $m > 0$ such that

$$(37) \quad |\bar{P}_p(\omega)| \leq \alpha L_\phi - m \quad (\omega \in \Omega).$$

Choose $\varepsilon_1 > 0$ such that $\hat{J}(z) \geq \hat{J}(\bar{z})$ whenever $\|z - \bar{z}\|_{\ell^1} \leq \varepsilon_1$. Put

$$B_{\bar{\omega}} := \max_{1 \leq n \leq N} \|K_p(\bar{\omega}_n)\|_{H^1(D)}.$$

Choose $\varepsilon > 0$ so that

$$\varepsilon \leq \varepsilon_1, \quad \varepsilon \leq \hat{z}_1, \quad B_p B_{\bar{\omega}} \varepsilon \leq \frac{m}{2}, \quad \frac{\alpha \gamma_1 \varepsilon}{2} \leq \frac{m}{4}.$$

A condition with zero left-hand side is void.

Let $\mu \in \mathcal{M}_p(\Omega)$ satisfy $\|\mu - \bar{\mu}\|_{\mathcal{M}_p(\Omega)} \leq \varepsilon$. There is nothing to prove if $J(\mu) = +\infty$. Otherwise, write $\xi := \mu_p$ and, for $1 \leq n \leq N$, set

$$z_n := \xi(\{\bar{\omega}_n\}), \quad z := (z_n)_{n=1}^N, \quad \xi_0 := \sum_{n=1}^N z_n \delta_{\bar{\omega}_n}, \quad \xi = \xi_0 + \tilde{\xi}.$$

Then $\tilde{\xi}$ has no mass at the support points, and the normalization isometry gives

$$\|z - \bar{z}\|_{\ell^1} + \|\tilde{\xi}\|_{\mathcal{M}(\Omega)} = \|\mu - \bar{\mu}\|_{\mathcal{M}_p(\Omega)} \leq \varepsilon.$$

Define

$$r' := \sum_{n=1}^N (z_n - \bar{z}_n) K_p(\bar{\omega}_n), \quad Q_p(\omega) := \langle r', K_p(\omega) \rangle_{H^1(D)}.$$

Then

$$\begin{aligned} \|r'\|_{H^1(D)} &\leq B_{\bar{\omega}} \|z - \bar{z}\|_{\ell^1} \leq B_{\bar{\omega}} \varepsilon, \\ |Q_p(\omega)| &\leq B_p B_{\bar{\omega}} \varepsilon \leq \frac{m}{2}. \end{aligned}$$

Combining this with (37) gives

$$(38) \quad |\bar{P}_p(\omega) + Q_p(\omega)| \leq \alpha L_\phi - \frac{m}{2} \quad (\omega \in \Omega).$$

The exact quadratic expansion of the fidelity in the direction $\tilde{\xi}$ and the nonnegativity of its quadratic remainder give

$$(39) \quad L(\mu) \geq \frac{1}{2} \left\| \sum_{n=1}^N z_n K_p(\bar{\omega}_n) - V \right\|_{H^1(D)}^2 + \int_{\Omega} (\bar{P}_p + Q_p) d\tilde{\xi}.$$

The measures ξ_0 and $\tilde{\xi}$ are mutually singular. Every atom of $\tilde{\xi}$ has magnitude at most $\|\tilde{\xi}\|_{\mathcal{M}(\Omega)} \leq \varepsilon \leq \hat{z}_1$. Moreover,

$$\Phi_\phi(\xi) = \sum_{n=1}^N \phi(|z_n|) + \Phi_\phi(\tilde{\xi}).$$

The local lower bound (35), applied to the atoms and combined with the definition on the continuous part, gives the linear term $L_\phi \|\tilde{\xi}\|_{\mathcal{M}(\Omega)}$ and an atomic remainder bounded below by

$$-\frac{\gamma_1}{2} \sum_{\omega \in \text{atom } \tilde{\xi}} |\tilde{\xi}(\{\omega\})|^2 \geq -\frac{\gamma_1 \varepsilon}{2} \sum_{\omega \in \text{atom } \tilde{\xi}} |\tilde{\xi}(\{\omega\})|,$$

because every atom magnitude is at most ε . Therefore

$$(40) \quad \Phi_\phi(\tilde{\xi}) \geq \left(L_\phi - \frac{\gamma_1 \varepsilon}{2} \right) \|\tilde{\xi}\|_{\mathcal{M}(\Omega)}.$$

Using (39), (38), and (40), we obtain

$$\begin{aligned} J(\mu) &\geq \widehat{J}(z) + \int_{\Omega} (\bar{P}_p + Q_p) d\tilde{\xi} + \alpha \Phi_{\phi}(\tilde{\xi}) \\ &\geq \widehat{J}(z) + \left(\frac{m}{2} - \frac{\alpha \gamma_1 \varepsilon}{2} \right) \|\tilde{\xi}\|_{\mathcal{M}(\Omega)} \\ &\geq \widehat{J}(\bar{z}) + \frac{m}{4} \|\tilde{\xi}\|_{\mathcal{M}(\Omega)} \geq J(\bar{\mu}). \end{aligned}$$

Thus $\bar{\mu}$ is an $\mathcal{M}_p(\Omega)$ -local minimizer. $\square$

Admissible activations. The two exponents in Assumption 3.1 play different roles: s_0 controls the value atom in $L^2(D)$ and is enough for Theorem 3.16; s_1 controls the full atom in $H^1(D)$ and is used by Theorems 3.17, 3.20, and 5.1. Since $\|K(\omega)\|_{L^2(D)} \leq \|K(\omega)\|_{H^1(D)}$, it is consistent to impose $s_0 \leq s_1$, as Assumption 3.1 does. The exponent s_1 is obtained by estimating $a\rho'(a \cdot x + b)$ separately. The condition $p > s_1$ normalizes the full H^1 atom; together with Assumption 3.4(2), it is sufficient for the finite-support conclusion in Theorem 3.20. The following exponents satisfy the pointwise bounds on a bounded domain.

activation	s_0	s_1	sufficient condition
$\tanh z$	0	1	$p > 1$
e^{-z^2}	0	1	$p > 1$
$\log(1 + e^z)$	1	1	$p > 1$
$(z\Phi(z))^2$	2	2	$p > 2$
$(z_+)^k$	k	k	$p > k$

For bounded activations such as $\tanh$ and the Gaussian, existence alone requires only $p > 0$; the condition $p > 1$ in the table is the one that also dominates the elementary global H^1 growth estimate. The fourth row is the squared Gaussian error linear unit, written GELU² below, with Φ the standard normal distribution function; since Φ takes values in $(0, 1)$ we have $|z\Phi(z)| \leq |z|$, and its derivative $2z\Phi(z)(\Phi(z) + z\varphi(z))$ grows linearly, which gives $s_0 = s_1 = 2$. The last row is listed for comparison with the positively homogeneous formulation of Section 4.

4. THE k -HOMOGENEOUS FORMULATION

The ReLU ^{k} activation function is of particular interest since it is widely used in the Sobolev training [14, 11]. For gradient fitting with ReLU ^{k} we use $k \geq 2$, so that $K(\omega) \in C^1(\overline{D})$. As we will see in this section, the induced penalty is $\psi_k(t) = t^q$ with $q = 2/(k+1) < 1$, so it has infinite right slope at zero. It does not extend the result in Section 3.

On the other hand, k -homogeneity makes the radial part of an inner parameter redundant: every atom can be moved onto the unit sphere $\mathbb{S}^d \subset \mathbb{R}^{d+1}$ at the cost of rescaling its outer coefficient. The parameter domain of the resulting problem is compact, so the dictionary is bounded, and neither the polynomial weight w_p nor the weighted space $\mathcal{M}_p(\Omega)$ of Section 3 is required. Throughout this section we work on $\mathcal{M}(\mathbb{S}^d)$ with the total-variation norm $\|\cdot\|_{\mathcal{M}(\mathbb{S}^d)}$, which we identify with the subspace of $\mathcal{M}(\Omega)$ of measures vanishing outside $\mathbb{S}^d$.

Assumption 4.1 (k -homogeneity). For some $k \geq 1$,

$$K(\lambda\omega) = \lambda^k K(\omega) \quad (\lambda > 0, \omega \in \mathbb{R}^{d+1}).$$

Lemma 4.2 (The ReLU ^{k} dictionary). *Let $k \geq 2$, let $D \subset \mathbb{R}^d$ be bounded with positive Lebesgue measure, let $\rho(z) = (z_+)^k$, and let*

$$[K(\omega)](x) = (a \cdot x + b)_+^k, \quad \omega = (a, b) \in \mathbb{R}^{d+1}.$$

Then

- (1) K satisfies Assumption 4.1 with degree k ;

- (2) $K(\omega) \in C^1(\mathbb{R}^d)$ with $\nabla K(\omega)(x) = k(a \cdot x + b)_+^{k-1}a$, and this classical gradient is also the distributional gradient of $K(\omega)$ on D ;
- (3) $K : \mathbb{S}^d \rightarrow H^1(D)$ is continuous; and
- (4) $C_S := \sup_{\omega \in \mathbb{S}^d} \|K(\omega)\|_{H^1(D)}$ satisfies $0 < C_S < \infty$.

Proof. (1) For $\lambda > 0$ one has $(\lambda z)_+ = \lambda z_+$, hence $(\lambda a \cdot x + \lambda b)_+^k = \lambda^k (a \cdot x + b)_+^k$ for every x .

(2) Set $h_k(z) := (z_+)^k$. On $(0, \infty)$ and on $(-\infty, 0)$ the function h_k is differentiable with $h'_k(z) = k(z_+)^{k-1}$. At $z = 0$ both one-sided difference quotients are $O(|z|^{k-1})$ and vanish in the limit because $k \geq 2$, so $h'_k(0) = 0 = k(0_+)^{k-1}$. The formula $h'_k(z) = k(z_+)^{k-1}$ therefore holds on all of $\mathbb{R}$, and h'_k is continuous because $k - 1 \geq 1$. The chain rule applied to $x \mapsto a \cdot x + b$ gives $\nabla K(\omega)(x) = k(a \cdot x + b)_+^{k-1}a$. Since D is bounded, both $K(\omega)$ and $\nabla K(\omega)$ are bounded and continuous on D ; a continuously differentiable function on a neighbourhood of D lies in $H^1(D)$ and its distributional gradient is its classical gradient.

(3) Let $\omega_j = (a_j, b_j) \rightarrow \omega = (a, b)$ in $\mathbb{S}^d$ and put $R_D := \sup_{x \in D} |x| < \infty$. Then

$$\sup_{x \in D} |(a_j \cdot x + b_j) - (a \cdot x + b)| \leq |a_j - a| R_D + |b_j - b| \rightarrow 0,$$

so the pre-activations converge uniformly on D . All of them lie in the compact interval $I := [-(R_D + 1), R_D + 1]$, because $|a_j \cdot x + b_j| \leq R_D + 1$ for every j and every $x \in D$ by $|a_j| \leq 1$, $|b_j| \leq 1$. The functions h_k and h'_k are uniformly continuous on I , whence

$$K(\omega_j) \rightarrow K(\omega) \quad \text{and} \quad h'_k(a_j \cdot x + b_j) \rightarrow h'_k(a \cdot x + b) \quad \text{uniformly on } D.$$

Since $\nabla K(\omega_j)(x) = h'_k(a_j \cdot x + b_j) a_j$, $a_j \rightarrow a$, and the scalar factors are uniformly bounded on D , the gradients converge uniformly on D as well. As D has finite Lebesgue measure, uniform convergence implies convergence in $L^2(D)$, so $K(\omega_j) \rightarrow K(\omega)$ in $H^1(D)$.

(4) Finiteness follows from (3) and compactness of $\mathbb{S}^d$, since a continuous real function attains a finite maximum on a compact set. For positivity take $\hat{\omega} = (0, 1) \in \mathbb{S}^d$, for which $[K(\hat{\omega})](x) = 1$ for every x ; hence $C_S \geq \|K(\hat{\omega})\|_{H^1(D)} = |D|^{1/2} > 0$. $\square$

In the remainder of the section $K : \mathbb{S}^d \rightarrow H^1(D)$ is continuous and Lemma 4.2 verifies this assumption and $C_S > 0$ for ReLU^k with $k \geq 2$, the range used for the gradient-augmented problem.

The following lemma extends the degree-one ReLU rescaling result [20, Appendix F, Proposition 3] to a k -homogeneous dictionary. We retain general positive exponents to make explicit how the homogeneity degree changes both the induced exponent and its coefficient.

Lemma 4.3 (Rescaling of a k -homogeneous network). *Let Assumption 4.1 hold, and let $\xi, \zeta > 0$. For a fixed nonzero represented coefficient $\tilde{c}$ and a unit direction $\hat{\omega} \in \mathbb{S}^d$ with $K(\hat{\omega}) \neq 0$, minimize*

$$\frac{|c|^\xi}{\xi} + \frac{|\omega|^\zeta}{\zeta}$$

over radial representations

$$\omega = \lambda \hat{\omega}, \quad \lambda > 0, \quad c K(\omega) = \tilde{c} K(\hat{\omega}).$$

The minimum equals

$$(41) \quad C_{k,\xi,\zeta} |\tilde{c}|^q, \quad q := \frac{\xi \zeta}{k\xi + \zeta}, \quad C_{k,\xi,\zeta} := \frac{1}{q} k^{-k\xi/(k\xi + \zeta)}.$$

Consequently, consider any finite representation without zero-function neurons and set

$$\hat{\omega}_n := \frac{\omega_n}{|\omega_n|}, \quad \tilde{c}_n := c_n |\omega_n|^k.$$

Among all independent radial rescalings that preserve each function $c_n K(\omega_n)$, the minimum total penalty is $C_{k,\xi,\zeta} \sum_n |\tilde{c}_n|^q$.

Proof. We reduce the rescaling constraint to a scalar minimization and solve it explicitly. Write $\omega = \lambda \hat{\omega}$, where $\lambda = |\omega| > 0$. Since $K(\hat{\omega}) \neq 0$, k -homogeneity makes the representation constraint equivalent to $c\lambda^k = \tilde{c}$. Hence, for fixed $\tilde{c}$, the radial penalty is

$$g(\lambda) = \frac{|\tilde{c}|^\xi}{\xi \lambda^{k\xi}} + \frac{\lambda^\zeta}{\zeta}.$$

It tends to $+\infty$ at both endpoints, and

$$g'(\lambda) = \lambda^{-k\xi-1} \left(\lambda^{k\xi+\zeta} - k|\tilde{c}|^\xi \right).$$

Thus g' changes sign exactly once, at

$$\lambda_* = (k|\tilde{c}|^\xi)^{1/(k\xi+\zeta)}.$$

It is therefore the unique global minimizer. The stationarity identity gives $|\tilde{c}|^\xi \lambda_*^{-k\xi} = \lambda_*^\zeta/k$; using the definition of q in (41), we obtain

$$\begin{aligned} g(\lambda_*) &= \left(\frac{1}{k\xi} + \frac{1}{\zeta} \right) \lambda_*^\zeta \\ &= \frac{1}{q} k^{-k\xi/(k\xi+\zeta)} |\tilde{c}|^q \\ &= C_{k,\xi,\zeta} |\tilde{c}|^q. \end{aligned}$$

For each normalized neuron $(\tilde{c}_n, \hat{\omega}_n)$, set

$$\lambda_n = (k|\tilde{c}_n|^\xi)^{1/(k\xi+\zeta)}, \quad \omega_n = \lambda_n \hat{\omega}_n, \quad c_n = \tilde{c}_n \lambda_n^{-k}.$$

By k -homogeneity, $c_n K(\omega_n) = \tilde{c}_n K(\hat{\omega}_n)$, and the displayed rescaling preserves this function. The radial variables are independent, so applying the scalar minimum to each neuron and summing proves the final assertion. $\square$

Remark 4.4 (The quadratic case and the meaning of α). For $\xi = \zeta = 2$,

$$q = \frac{2}{k+1}, \quad C_k := C_{k,2,2} = \frac{k+1}{2} k^{-k/(k+1)}.$$

In particular $C_1 = 1$ and $C_2 = \frac{3}{2} 2^{-2/3}$. Since C_k depends on k , equal nominal values of α in the penalty $\alpha \sum_n |c_n|^q$ for different powers k do *not* correspond to equal coefficients of the original separable penalty.

For $k \geq 2$ define the scalar penalty

$$\psi_k(t) := t^q, \quad q := \frac{2}{k+1} \in (0, 1), \quad t \geq 0,$$

which is the exponent of Lemma 4.3 for $\xi = \zeta = 2$. It is continuous and strictly increasing with $\psi_k(0) = 0$ and $\psi'_k(0^+) = +\infty$, and it is subadditive:

$$(42) \quad \psi_k(s+t) \leq \psi_k(s) + \psi_k(t) \quad (s, t \geq 0).$$

Indeed, for $s+t > 0$ the numbers $u := s/(s+t)$ and $1-u = t/(s+t)$ lie in $[0, 1]$, where $v^q \geq v$; hence $\psi_k(s) + \psi_k(t) = (s+t)^q (u^q + (1-u)^q) \geq (s+t)^q$. Iterating (42) and passing to the limit in the finite partial sums gives, for every summable sequence $(t_j)_{j \geq 1}$ of nonnegative numbers,

$$(43) \quad \psi_k\left(\sum_{j \geq 1} t_j\right) \leq \sum_{j \geq 1} \psi_k(t_j).$$

Definition 4.5 (Homogeneous measure penalty). For $\mu \in \mathcal{M}(\mathbb{S}^d)$ with atomic and continuous parts μ_{atom} and μ_{cont} as in Section 2.1, the atom set is at most countable. Hence the following sum of nonnegative terms is well defined in $[0, +\infty]$ and independent of its enumeration. Set

$$\Phi_{\psi_k}(\mu) := \begin{cases} \sum_{\omega \in \text{atom } \mu} \psi_k(|\mu(\{\omega\})|), & \mu_{\text{cont}} = 0, \\ +\infty, & \mu_{\text{cont}} \neq 0. \end{cases}$$

The functional is nonnegative, vanishes only at the zero measure, and, whenever $\Phi_{\psi_k}(\mu) < \infty$, countable subadditivity gives

$$(44) \quad \Phi_{\psi_k}(\mu) \geq \psi_k(\|\mu\|_{\mathcal{M}(\mathbb{S}^d)}) = \|\mu\|_{\mathcal{M}(\mathbb{S}^d)}^q.$$

For a finite representation $\mu = \sum_{n=1}^N c_n \delta_{\omega_n}$, repeated locations are merged. The triangle inequality and (42) show that merging cannot increase $\sum_n \psi_k(|c_n|)$.

4.1. The measure problem on the sphere. For $\mu \in \mathcal{M}(\mathbb{S}^d)$ we now set

$$(45) \quad J_{\psi_k}(\mu) := L(\mu) + \alpha \Phi_{\psi_k}(\mu),$$

$$(46) \quad J_{\psi_k}^M(\mu) := l^M(\mu) + \alpha \Phi_{\psi_k}(\mu),$$

with the convention that a sum containing $+\infty$ equals $+\infty$. Since the empirical fidelity is also one half of a squared Hilbert-space norm and its dictionary is continuous on the compact sphere, all results below hold for the empirical objective as well. The empirical fidelity (6) was introduced for finite networks; for ReLU^k with $k \geq 2$ it is defined on all of $\{\mu \in \mathcal{M}(\mathbb{S}^d) : \Phi_{\psi_k}(\mu) < \infty\}$, provided the sampled data $V(x^m)$ and g^m are given. Indeed, such a μ is purely atomic with $\sum_j |c_j| < \infty$, while Lemma 4.2(2)–(3) and compactness of $\mathbb{S}^d$ bound $|[K(\omega)](x^m)|$ and $|\nabla K(\omega)(x^m)|$ uniformly in $\omega \in \mathbb{S}^d$ and m ; hence $\sum_j c_j [K(\omega_j)](x^m)$ and $\sum_j c_j \nabla K(\omega_j)(x^m)$ converge absolutely, so both residuals in $l^M(\mu)$ are defined. We study the minimization problem

$$(P_{\psi_k}) \quad \min_{\mu \in \mathcal{M}(\mathbb{S}^d)} J_{\psi_k}(\mu).$$

For a finite network with distinct locations $\omega_n \in \mathbb{S}^d$, the definition of Φ_{ψ_k} gives

$$J_{\psi_k}(\mu_{\omega,c}) = L(\mu_{\omega,c}) + \alpha \sum_{n=1}^N |c_n|^q,$$

which is the objective minimized by the finite-dimensional solver in Algorithm 2; the empirical objective (46) restricts in the same way.

Local minimizers of (P_{ψ_k}) are understood as in Definition 3.14, with $\mathcal{M}_p(\Omega)$ replaced by $\mathcal{M}(\mathbb{S}^d)$, J by J_{ψ_k} , and the weighted measure norm by the total-variation norm.

Proposition 4.6 (Finite truncations and equality of the infima). *Let $V \in H^1(D)$, let $K : \mathbb{S}^d \rightarrow H^1(D)$ be continuous with bound C_S , let $\mu \in \mathcal{M}(\mathbb{S}^d)$ satisfy $\Phi_{\psi_k}(\mu) < \infty$, let $(\omega_j)_{j \in \mathcal{I}}$, $\mathcal{I} \subseteq \mathbb{N}$, enumerate atom μ , write $c_j := \mu(\{\omega_j\}) \neq 0$, and set*

$$\mu_m := \sum_{j \in \mathcal{I}, j \leq m} c_j \delta_{\omega_j}.$$

Then

$$\mathcal{N}\mu \in H^1(D), \quad \|\mathcal{N}\mu\|_{H^1(D)} \leq C_S \|\mu\|_{\mathcal{M}(\mathbb{S}^d)},$$

$$\|\mathcal{N}\mu_m - \mathcal{N}\mu\|_{H^1(D)} \leq C_S \|\mu_m - \mu\|_{\mathcal{M}(\mathbb{S}^d)} \longrightarrow 0.$$

Moreover, $J_{\psi_k}(\mu_m) \rightarrow J_{\psi_k}(\mu)$. Consequently the all-measures infimum and the finite-network infimum coincide:

$$(47) \quad \inf_{\mu \in \mathcal{M}(\mathbb{S}^d)} J_{\psi_k}(\mu) = \inf_{N \in \mathbb{N}_0} \inf_{\substack{\omega_n \in \mathbb{S}^d, \\ n=1, \dots, N}} \inf_{c_n \in \mathbb{R}} \left[L(\mu_{\omega,c}) + \alpha \sum_{n=1}^N \psi_k(|c_n|) \right].$$

Proof. Since $\Phi_{\psi_k}(\mu) < \infty$, Definition 4.5 gives $\mu_{\text{cont}} = 0$, so $\mu = \sum_{j \in \mathcal{I}} c_j \delta_{\omega_j}$ with distinct ω_j and $c_j \neq 0$. Since $\sum_{j \in \mathcal{I}} |c_j| = \|\mu\|_{\mathcal{M}(\mathbb{S}^d)} < \infty$, the tail identity

$$\|\mu_m - \mu\|_{\mathcal{M}(\mathbb{S}^d)} = \sum_{j \in \mathcal{I}, j > m} |c_j| \longrightarrow 0.$$

Bounded linearity of $\mathcal{N}$ therefore gives $\mathcal{N}\mu \in H^1(D)$ and the stated norm bounds. Continuity of $u \mapsto \frac{1}{2}\|u - V\|_{H^1(D)}^2$ gives $L(\mu_m) \rightarrow L(\mu)$, while

$$\Phi_{\psi_k}(\mu_m) = \sum_{j \in \mathcal{I}, j \leq m} \psi_k(|c_j|) \longrightarrow \Phi_{\psi_k}(\mu)$$

by monotone convergence. Combining these two limits proves $J_{\psi_k}(\mu_m) \rightarrow J_{\psi_k}(\mu)$.

It remains to compare the infima; call them i_{meas} and i_{disc} , in the order in which they appear in (47). Every finite network defines an admissible measure, and its finite-dimensional objective equals the corresponding measure objective. Hence $i_{\text{meas}} \leq i_{\text{disc}}$. Conversely, let μ satisfy $J_{\psi_k}(\mu) < \infty$. Its truncations μ_m are finite networks and satisfy $J_{\psi_k}(\mu_m) \rightarrow J_{\psi_k}(\mu)$. Therefore $i_{\text{disc}} \leq J_{\psi_k}(\mu)$. Taking the infimum over such μ gives $i_{\text{disc}} \leq i_{\text{meas}}$. $\square$

Proposition 4.6 shows that a measure of finite penalty automatically has finite fidelity. Since conversely $J_{\psi_k}(\mu) < \infty$ contains $\alpha \Phi_{\psi_k}(\mu) < \infty$, the two conditions agree:

$$J_{\psi_k}(\mu) < \infty \iff \Phi_{\psi_k}(\mu) < \infty \quad (\mu \in \mathcal{M}(\mathbb{S}^d)),$$

and either may be used as the standing finiteness hypothesis.

Proposition 4.6 is used only as a bridge between the finite-network problem and (P_{ψ_k}) . It equates the two infima; attainment is established in Theorem 4.9 from the finite-dimensional argument and is not claimed here.

Theorem 4.7. *Let $k \geq 2$, $q = 2/(k+1) \in (0, 1)$, $\alpha > 0$, $V \in H^1(D)$, and let $K : \mathbb{S}^d \rightarrow H^1(D)$ be continuous. Let $\bar{\mu} \in \mathcal{M}(\mathbb{S}^d)$ satisfy $J_{\psi_k}(\bar{\mu}) < \infty$ and be optimal against perturbations of the weights of its own atoms: there is $\varepsilon > 0$ such that*

$$(48) \quad J_{\psi_k}(\bar{\mu}) \leq J_{\psi_k}(\bar{\mu} + t\delta_\omega) \quad \text{for every } \omega \in \text{atom } \bar{\mu} \text{ and every } |t| < \varepsilon.$$

Set

$$B(\bar{\mu}) := \|P_{\bar{\mu}}\|_{C(\mathbb{S}^d)}.$$

Then $\bar{\mu}$ is purely atomic, $B(\bar{\mu}) < \infty$, and every atom $\omega_n \in \text{atom } \bar{\mu}$ with weight $c_n := \bar{\mu}(\{\omega_n\})$ satisfies

$$(49) \quad P_{\bar{\mu}}(\omega_n) + \alpha q |c_n|^{q-1} \text{sign}(c_n) = 0.$$

If $B(\bar{\mu}) = 0$, then $\bar{\mu} = 0$. If $B(\bar{\mu}) > 0$, then

$$(50) \quad |c_n| \geq c_{\min}(\bar{\mu}) := \left(\frac{\alpha q}{B(\bar{\mu})} \right)^{1/(1-q)} > 0 \quad \text{for every } \omega_n \in \text{atom } \bar{\mu}.$$

Hypothesis (48) holds in particular if $\bar{\mu}$ is a local minimizer of (P_{ψ_k}) in total variation in the sense specified above.

Proof. We derive (49) from a two-sided perturbation of a single atom weight, and then convert it into the lower bound (50).

Since $J_{\psi_k}(\bar{\mu}) < \infty$ we have $\Phi_{\psi_k}(\bar{\mu}) < \infty$, so $\bar{\mu}_{\text{cont}} = 0$ and $\bar{\mu}$ is purely atomic. By Proposition 4.6, applied to $\bar{\mu}$, $\mathcal{N}\bar{\mu} \in H^1(D)$, so $r_{\bar{\mu}} \in H^1(D)$ is well defined. The map $P_{\bar{\mu}}$ is continuous on $\mathbb{S}^d$, being the composition of the continuous map K with the continuous linear functional $\langle r_{\bar{\mu}}, \cdot \rangle_{H^1(D)}$, and $\mathbb{S}^d$ is compact, so $B(\bar{\mu}) < \infty$.

Fix $\omega_n \in \text{atom } \bar{\mu}$, write $c_n = \bar{\mu}(\{\omega_n\}) \neq 0$, let $s \in \{-1, 1\}$, and consider $\mu_t := \bar{\mu} + ts\delta_{\omega_n}$ for $0 < t < \min\{\varepsilon, |c_n|\}$. The measure μ_t differs from $\bar{\mu}$ only in the weight of the atom ω_n , which becomes $c_n + ts \neq 0$, so $\text{atom } \mu_t = \text{atom } \bar{\mu}$ and

$$\Phi_{\psi_k}(\mu_t) - \Phi_{\psi_k}(\bar{\mu}) = \psi_k(|c_n + ts|) - \psi_k(|c_n|),$$

all other summands being unchanged. For the fidelity, $\mathcal{N}\mu_t = \mathcal{N}\bar{\mu} + tsK(\omega_n)$, so the residual is $r_{\bar{\mu}} + tsK(\omega_n)$ and expanding the square gives the exact identity

$$\begin{aligned} L(\mu_t) - L(\bar{\mu}) &= ts\langle r_{\bar{\mu}}, K(\omega_n) \rangle_{H^1(D)} + \frac{t^2}{2} \|K(\omega_n)\|_{H^1(D)}^2 \\ &= tsP_{\bar{\mu}}(\omega_n) + \frac{t^2}{2} \|K(\omega_n)\|_{H^1(D)}^2. \end{aligned}$$

Because $c_n \neq 0$, the map $z \mapsto |z|^q$ is differentiable at c_n with derivative $q|c_n|^{q-1} \text{sign}(c_n)$. Hypothesis (48) gives $J_{\psi_k}(\mu_t) - J_{\psi_k}(\bar{\mu}) \geq 0$; dividing by $t > 0$ and letting $t \downarrow 0$ yields

$$0 \leq s \left[P_{\bar{\mu}}(\omega_n) + \alpha q |c_n|^{q-1} \text{sign}(c_n) \right].$$

Choosing $s = 1$ and $s = -1$ gives (49).

Suppose $B(\bar{\mu}) = 0$, that is, $P_{\bar{\mu}} \equiv 0$ on $\mathbb{S}^d$. If $\bar{\mu}$ had an atom ω_n , then (49) would read $\alpha q |c_n|^{q-1} \text{sign}(c_n) = 0$, which is impossible because $\alpha > 0$, $q > 0$, and $|c_n|^{q-1} \in (0, \infty)$ for $c_n \neq 0$. Hence $\text{atom } \bar{\mu} = \emptyset$, and since $\bar{\mu}$ is purely atomic, $\bar{\mu} = 0$.

Suppose $B(\bar{\mu}) > 0$. Taking absolute values in (49) gives

$$\alpha q |c_n|^{q-1} = |P_{\bar{\mu}}(\omega_n)| \leq B(\bar{\mu}).$$

Since $q - 1 = -(1 - q) < 0$, this reads $|c_n|^{-(1-q)} \leq B(\bar{\mu})/(\alpha q)$, hence $|c_n|^{1-q} \geq \alpha q/B(\bar{\mu})$; raising both sides to the positive power $1/(1 - q)$ preserves the inequality and gives (50). $\square$

Remark 4.8. The identity $\psi'_k(0^+) = +\infty$ has the following consequences.

- (1) Inserting a new atom at $\omega \notin \text{atom } \bar{\mu}$ with weight ts , $s \in \{-1, 1\}$, changes the penalty by αt^q . Since $t^q/t = t^{q-1} \rightarrow \infty$ as $t \downarrow 0$, division by t produces no finite limit, and hence no pointwise condition on $P_{\bar{\mu}}(\omega)$ away from $\text{atom } \bar{\mu}$. Theorem 4.7 is restricted to existing atoms for this reason.
- (2) The zero measure is a local minimizer of (P_{ψ_k}) in total variation. Indeed, $J_{\psi_k}(\mu) = \infty$ whenever $\Phi_{\psi_k}(\mu) = \infty$, while for μ with $\Phi_{\psi_k}(\mu) < \infty$ and $\|\mu\|_{\mathcal{M}(\mathbb{S}^d)} = \varepsilon$,

$$\begin{aligned} J_{\psi_k}(\mu) - J_{\psi_k}(0) &= \langle -V, \mathcal{N}\mu \rangle_{H^1(D)} + \frac{1}{2} \|\mathcal{N}\mu\|_{H^1(D)}^2 + \alpha \Phi_{\psi_k}(\mu) \\ &\geq -\|V\|_{H^1(D)} C_S \varepsilon + \alpha \varepsilon^q, \end{aligned}$$

by (44), the bound $\|\mathcal{N}\mu\|_{H^1(D)} \leq C_S \varepsilon$ of Proposition 4.6, and the Cauchy–Schwarz inequality. Since $q < 1$, the right-hand side is positive for every sufficiently small $\varepsilon > 0$.

An insertion criterion based on an infinitesimal weight perturbation is therefore unavailable for this penalty, and Algorithm 2 uses a finite-step decrease test instead.

4.2. Global existence and a uniform width bound.

Theorem 4.9 (Existence and width bound). *Let $k \geq 2$, $q = 2/(k + 1) \in (0, 1)$, $\alpha > 0$, let $V \in H^1(D)$ with $V \neq 0$, and let $K : \mathbb{S}^d \rightarrow H^1(D)$ be continuous with $C_S = \sup_{\omega \in \mathbb{S}^d} \|K(\omega)\|_{H^1(D)} > 0$; the latter holds for ReLU^k by Lemma 4.2(4). Then (P_{ψ_k}) admits a global minimizer, and its minimum value equals the finite-network infimum on the right-hand side of (47). Every global minimizer $\bar{\mu}$ is a finite atomic measure satisfying*

$$\# \text{ atom } \bar{\mu} \leq N_{\max} := \frac{\|V\|_{H^1(D)}^2}{2\alpha c_{\min, \text{glob}}^q}, \quad c_{\min, \text{glob}} := \left(\frac{\alpha q}{\|V\|_{H^1(D)} C_S} \right)^{1/(1-q)}.$$

Remark 4.10. The excluded cases are immediate. If $V = 0$, then $\mu = 0$ is the unique global minimizer by the nonnegativity of the fidelity and the strict positivity of Φ_{ψ_k} away from zero. If $C_S = 0$, then $K \equiv 0$, so the fidelity is constant and the same penalty argument again makes $\mu = 0$ the unique global minimizer. Thus the atom bound is zero in both cases. The assumptions $V \neq 0$ and $C_S > 0$ are needed only to ensure that $c_{\min, \text{glob}} \in (0, \infty)$ and $N_{\max} < \infty$ are well defined.

Proof. We first obtain minimizers at each fixed width by compactness, then derive one width estimate for both the existence argument and arbitrary global minimizers.

For $N \in \mathbb{N}_0$, set

$$F_N(\omega, c) := L(\mu_{\omega, c}) + \alpha \sum_{n=1}^N |c_n|^q, \quad m_N := \inf F_N,$$

with $F_0 = m_0 = \frac{1}{2} \|V\|_{H^1(D)}^2$. For $N \geq 1$, continuity of K , scalar multiplication, and finite addition show that $(\omega, c) \mapsto \sum_{n=1}^N c_n K(\omega_n)$ is continuous into $H^1(D)$; hence F_N is continuous. Any minimizing sequence may be chosen in the sublevel $F_N \leq m_0 + 1$. Since the fidelity is nonnegative, this sublevel satisfies

$$|c_n| \leq R := \left(\frac{m_0 + 1}{\alpha} \right)^{1/q} \quad (n = 1, \dots, N).$$

It therefore lies in the compact set $(\mathbb{S}^d)^N \times [-R, R]^N$. Compactness gives a convergent subsequence, and continuity shows that its limit attains m_N . The case $N = 0$ is immediate, and we have $m_{N+1} \leq m_N$.

We use the following estimate twice. Suppose that a measure μ satisfies $J_{\psi_k}(\mu) \leq m_0$ and the atomwise optimality hypothesis (48). Theorem 4.7 shows that μ is purely atomic. The nonnegativity of the penalty gives $\|r_\mu\|_{H^1(D)} \leq \|V\|_{H^1(D)}$, and hence

$$B(\mu) \leq \|V\|_{H^1(D)} C_S.$$

If $\mu \neq 0$, Theorem 4.7 gives $B(\mu) > 0$ and

$$|\mu(\{\omega\})| \geq \left(\frac{\alpha q}{B(\mu)} \right)^{1/(1-q)} \geq c_{\min, \text{glob}} \quad (\omega \in \text{atom } \mu).$$

Consequently, for every finite $A \subseteq \text{atom } \mu$,

$$\#A \alpha c_{\min, \text{glob}}^q \leq \alpha \sum_{\omega \in A} |\mu(\{\omega\})|^q \leq \alpha \Phi_{\psi_k}(\mu) \leq J_{\psi_k}(\mu) \leq \frac{1}{2} \|V\|_{H^1(D)}^2.$$

An infinite atom set would contain finite subsets of arbitrarily large cardinality. Thus $\text{atom } \mu$ is finite and its cardinality is at most $N_{\max}$; the same conclusion is immediate when $\mu = 0$.

Set $N_{\text{cut}} := \lfloor N_{\max} \rfloor$. The estimate above applies to every minimizer defining m_N , since each coefficient can be varied independently. Thus, for $N \geq N_{\text{cut}}$, that minimizer has width at most N_{cut} , so $m_{N_{\text{cut}}} \leq m_N$. The reverse inequality follows from the monotonicity of (m_N) . Hence

$$\inf_{N \in \mathbb{N}_0} m_N = m_{N_{\text{cut}}}.$$

Since $m_{N_{\text{cut}}}$ is attained, Proposition 4.6 gives a global minimizer of (P_{ψ_k}) with the same value. Every global minimizer satisfies the hypotheses of the estimate above and therefore has at most $N_{\max}$ atoms. $\square$

Corollary 4.11 (Finite support of total-variation local minimizers). *Let the hypotheses of Theorem 4.7 on k, q, α, V, K hold and let $\bar{\mu} \neq 0$ be a local minimizer of (P_{ψ_k}) in total variation. Then $B(\bar{\mu}) > 0$, the measure $\bar{\mu}$ is finite atomic, and*

$$\# \text{atom } \bar{\mu} \leq \frac{J_{\psi_k}(\bar{\mu})}{\alpha c_{\min}(\bar{\mu})^q}.$$

Proof. By Theorem 4.7, $B(\bar{\mu}) = 0$ would force $\bar{\mu} = 0$; hence $B(\bar{\mu}) > 0$ and every atom weight satisfies $|c_n| \geq c_{\min}(\bar{\mu}) > 0$. Since ψ_k is nondecreasing and the fidelity is nonnegative,

$$N\alpha c_{\min}(\bar{\mu})^q \leq \alpha \Phi_{\psi_k}(\bar{\mu}) \leq J_{\psi_k}(\bar{\mu}) < \infty$$

with $N = \# \text{ atom } \bar{\mu} \in \mathbb{N}_0 \cup \{\infty\}$, which forces $N < \infty$ and gives the stated bound. $\square$

The estimate of Corollary 4.11 depends on the minimizer through both $J_{\psi_k}(\bar{\mu})$ and $B(\bar{\mu})$ and is therefore an a posteriori bound. It is not computable in advance, unlike the a priori bound $N_{\max}$ of Theorem 4.9, which is uniform over all global minimizers and computable from $\|V\|_{H^1(D)}$, C_S , α and q alone. On a global minimizer the two are comparable and the a posteriori bound is the sharper one, since $J_{\psi_k}(\bar{\mu}) \leq \frac{1}{2}\|V\|_{H^1(D)}^2$ and $c_{\min}(\bar{\mu}) \geq c_{\min, \text{glob}}$ there.

5. ALGORITHMS

This section describes the computational pipeline: the training data are generated via the open-loop problem, candidate neurons are inserted greedily, and different insertion methods are chosen given the problem setting.

Data generation via the open-loop problem. Set $U := L^2(t_0, T; \mathbb{R}^{m_u})$ and assume that for every $u \in U$ the state equation in (P) has a unique solution $y(u)$ on $[t_0, T]$ and that the control-to-state map $u \mapsto y(u)$ is continuously Fréchet differentiable from U into $C([t_0, T]; \mathbb{R}^d)$. The objective of (P) is then the reduced functional $u \mapsto \mathcal{C}(u; t_0, x)$ on U , and the standard adjoint (Lagrangian) calculus — see, e.g., [12] — gives its gradient

$$\nabla_u \mathcal{C}(u; t_0, x) = 2\eta u + g(y(u))^\top p \quad \text{in } U,$$

where p solves the costate equation of (TPBVP) backward in time along $(y(u), u)$ in place of (y^*, u^*) . Each gradient evaluation therefore costs one forward solve of the state equation and one backward solve of the costate equation, and the implicit derivative of $u \mapsto y(u)$ is never formed. For the data-driven approach it is essential that the initial dataset $\{x^j, V(0, x^j), g^j\}_{j=1}^M$ is *causality-free*: the samples are generated without approximate values at neighboring points. [13] surveys generation methods. As in [1], we minimize $\mathcal{C}(\cdot; t_0, x)$ with Barzilai–Borwein step sizes. The converged initial costate supplies g^j , subject to that reservation and the differentiability qualification of Section 1.

The two algorithms. We use two algorithms of primal–dual active point (PDAP) type, in the sense of the accelerated conditional-gradient method for measure spaces of [21]. Algorithm 1 follows the nonconvex variant of [20], whose scalar penalty satisfies $\phi'(0) < \infty$ as in Assumption 3.2; Algorithm 2 is an adaptation of it to the penalty ψ_k , whose infinite slope at the origin is explicitly excluded by that reference.

In both algorithms the coefficient-correction step freezes the locations ω and applies a semismooth Newton method to the resulting finite-dimensional coefficient problem. They are specified in their respective subsections below. In either case the method is local and its output is retained only when it does not increase the post-insertion objective.

For the sampled fidelity, write $P_\mu^M(\omega) := Dl^M(\mu)[\delta_\omega]$, and let $\|K(\omega)\|_M$ denote the sampled value–gradient norm associated with l^M .

For a joint insertion $\sum_{i=1}^T c_i \delta_{\omega_i}$ at distinct new locations the penalty is additive, but the fidelity change contains pairwise cross terms between the inserted features, which no individual test controls. For $N_{\text{ins}} = 1$ the individual decrease statement applies to every accepted candidate, while Corollary 5.2 additionally requires an exact global maximizer. For $N_{\text{ins}} > 1$ the individual estimates do not compose, and neither a decrease per outer iteration nor monotonicity of the objective along the iterates is claimed. Restoring a batch decrease guarantee requires recomputing the residual after each insertion, or testing the joint increment of the whole batch.

5.1. Algorithm 1: the log penalty on the normalized measure.

Theorem 5.1. *Suppose that Assumption 3.1 holds, $p > s_1$, and $\alpha > 0$. Let ϕ satisfy Assumptions 3.2 and 3.3. For $\mu \in \mathcal{M}_p(\Omega)$ define*

$$\begin{aligned}\Delta(\mu, \omega) &:= \max\{|P_{p,\mu}(\omega)| - \alpha L_\phi, 0\}, \\ \Delta(\mu) &:= \sup_{\omega \in \Omega} \Delta(\mu, \omega) = \max\{\|P_{p,\mu}\|_\infty - \alpha L_\phi, 0\}.\end{aligned}$$

For a finite atomic μ , define

$$(51) \quad R(\mu) := \max \left\{ 1, \left(\frac{2^{s_1+1} C_K \|r_\mu\|_{H^1(D)}}{\alpha L_\phi} \right)^{1/(p-s_1)} \right\}.$$

Every ω with $\Delta(\mu, \omega) > 0$ belongs to $\overline{B}_{R(\mu)}$. If $\Delta(\mu) > 0$, then $B_p > 0$ and $|P_{p,\mu}|$ attains its maximum at some $\omega^* \in \overline{B}_{R(\mu)}$, with $\Delta(\mu, \omega^*) = \Delta(\mu)$. For every ω with $\Delta(\mu, \omega) > 0$, the insertion

$$(52) \quad \kappa(\omega) := -\frac{\Delta(\mu, \omega)}{B_p^2} \text{sign}(P_{p,\mu}(\omega)), \quad c(\omega) := \frac{\kappa(\omega)}{w_p(\omega)},$$

satisfies

$$(53) \quad J(\mu + c(\omega)\delta_\omega) \leq J(\mu) - \frac{\Delta(\mu, \omega)^2}{2B_p^2}.$$

Proof. We first establish attainment of the certificate maximum $\|P_{p,\mu}\|_\infty$ when $\Delta(\mu) > 0$. $p > s_1$ and (4) give $K_p \in C_0(\Omega; H^1(D))$, hence $B_p < \infty$ and $P_{p,\mu} \in C_0(\Omega)$. If $\Delta(\mu) > 0$, then $\|P_{p,\mu}\|_\infty > \alpha L_\phi > 0$. Choose a closed ball outside which $|P_{p,\mu}| < \|P_{p,\mu}\|_\infty/2$; this is possible because $P_{p,\mu}$ vanishes at infinity. The continuous function $|P_{p,\mu}|$ attains its maximum on that compact ball. Moreover, $B_p = 0$ would imply $P_{p,\mu} = 0$, so $B_p > 0$.

We now bound the change caused by an arbitrary normalized insertion $\kappa\delta_\omega$ with $\kappa \in \mathbb{R}$. The explicit atomic-continuous representation of Φ_ϕ , the monotonicity and subadditivity of ϕ , and Lemma 3.6 give

$$(54) \quad \begin{aligned} \Phi_\phi(\mu_p + \kappa\delta_\omega) - \Phi_\phi(\mu_p) &\leq \phi(|\kappa|) && \text{by subadditivity at the point } \omega \\ &\leq L_\phi |\kappa| && \text{by the tangent bound at zero.} \end{aligned}$$

Since $(\mu + \kappa w_p(\omega)^{-1}\delta_\omega)_p = \mu_p + \kappa\delta_\omega$, the quadratic fidelity has the exact expansion

$$(55) \quad \begin{aligned} J\left(\mu + \frac{\kappa}{w_p(\omega)}\delta_\omega\right) - J(\mu) &\leq \kappa P_{p,\mu}(\omega) + \frac{\kappa^2}{2} \|K_p(\omega)\|_{H^1(D)}^2 + \alpha L_\phi |\kappa| && \text{by (54)} \\ &\leq \kappa P_{p,\mu}(\omega) + \frac{B_p^2 \kappa^2}{2} + \alpha L_\phi |\kappa| && \text{by the definition of } B_p. \end{aligned}$$

Fix $\omega \in \Omega$ with $\Delta(\mu, \omega) > 0$. Then $|P_{p,\mu}(\omega)| = \alpha L_\phi + \Delta(\mu, \omega)$, and substitution of $\kappa = \kappa(\omega)$ from (52) into (55) yields

$$\begin{aligned} J(\mu + c(\omega)\delta_\omega) - J(\mu) &\leq -|\kappa(\omega)|(\alpha L_\phi + \Delta(\mu, \omega)) + \frac{B_p^2 |\kappa(\omega)|^2}{2} + \alpha L_\phi |\kappa(\omega)| && \text{since } \text{sign } \kappa(\omega) = -\text{sign } P_{p,\mu}(\omega) \\ &= -|\kappa(\omega)|\Delta(\mu, \omega) + \frac{B_p^2 |\kappa(\omega)|^2}{2} && \text{the threshold terms cancel} \\ &= -\frac{\Delta(\mu, \omega)^2}{2B_p^2} && \text{by the choice of } \kappa(\omega). \end{aligned}$$

This proves (53); at $\omega = \omega^*$ it gives the decrease $\Delta(\mu)^2/(2B_p^2)$.

Finally, we derive the radius from (4). If $|\omega| \geq 1$, then Cauchy–Schwarz gives

$$\begin{aligned} |P_{p,\mu}(\omega)| &\leq \|r_\mu\|_{H^1(D)} \|K_p(\omega)\|_{H^1(D)} && \text{by Cauchy–Schwarz} \\ &\leq C_K \|r_\mu\|_{H^1(D)} \frac{(1+|\omega|)^{s_1}}{1+|\omega|^p} && \text{by (4)} \\ &\leq 2^{s_1} C_K \|r_\mu\|_{H^1(D)} |\omega|^{-(p-s_1)} && \text{since } |\omega| \geq 1. \end{aligned}$$

If $|\omega| > R(\mu)$, the last expression is strictly smaller than $\alpha L_\phi/2$ by (51). This proves the claimed localization of every accepted candidate and every maximizer. $\square$

Corollary 5.2. *Let $(\mu_k)_{k \geq 0}$ be finite atomic iterates with $J(\mu_0) < \infty$, and write $\Delta_k := \Delta(\mu_k)$. If $\Delta_k > 0$, insert at an exact maximizer given by Theorem 5.1. Then correct the coefficients, accepting the corrected iterate only if it does not increase the objective. Then*

$$\begin{aligned} \sum_{k=0}^{\infty} \Delta_k^2 &\leq 2B_p^2 J(\mu_0), \\ \Delta_k &\longrightarrow 0, \\ \min_{0 \leq k < n} \Delta_k &\leq B_p \sqrt{\frac{2J(\mu_0)}{n}} \quad (n \geq 1). \end{aligned}$$

These conclusions do not assert convergence of (μ_k) to a global minimizer or an objective-gap rate.

Proof. If $B_p = 0$, then $\Delta_k = 0$ for every k and the claims are immediate. Otherwise Theorem 5.1 and the correction hypothesis give, for every $n \geq 1$,

$$\sum_{k=0}^{n-1} \frac{\Delta_k^2}{2B_p^2} \leq J(\mu_0) - J(\mu_n) \leq J(\mu_0).$$

Letting $n \rightarrow \infty$ gives the first claim, and summability gives $\Delta_k \rightarrow 0$. Finally,

$$\min_{0 \leq k < n} \Delta_k^2 \leq \frac{1}{n} \sum_{k=0}^{n-1} \Delta_k^2 \leq \frac{2B_p^2 J(\mu_0)}{n},$$

which proves the stated rate. $\square$

For the sampled data, Algorithm 1 minimizes

$$(56) \quad J^M(\omega, c) := l^M(\mu_{\omega, c}) + \alpha \sum_{n=1}^N \phi_\gamma(w_p(\omega_n) | c_n|).$$

Its normalized insertion profile is

$$P_{p, \mu_t}^M(\omega) := \frac{P_{\mu_t}^M(\omega)}{w_p(\omega)}.$$

For fixed nodes, introduce the normalized coefficients $u_n := w_p(\omega_n) c_n$. The outer-weight correction then minimizes

$$l^M \left(\sum_{n=1}^N \frac{u_n}{w_p(\omega_n)} \delta_{\omega_n} \right) + \alpha \sum_{n=1}^N \phi_\gamma(|u_n|).$$

Theorem 5.1 and Corollary 5.2 apply verbatim with the sampled value–gradient inner product in place of the $H^1(D)$ inner product. The empirical search radius is therefore given by (51), using the sampled residual norm and the corresponding sampled feature-growth constants.

The insertion step. The theorem requires finding a maximizer of $|P_{p,\mu_t}^M(\omega)|$. Finding a global maximizer is generally intractable; for ReLU dictionaries it is NP-hard [2, Section 3.3]. The practical multistart search is proposed in the PDAP algorithm in the paper [20]. At the current iterate $\mu_t = \mu_{\omega^{(t)}, c^{(t)}}$ a candidate ω is therefore accepted when

$$(57) \quad \Delta(\mu_t, \omega) > 0, \quad \text{that is,} \quad |P_{p,\mu_t}^M(\omega)| > \alpha L_\phi.$$

Theorem 5.1 provides two additional pieces of information. First, it gives the bound $R(\mu_t)$ for sampling the initial points at each iteration. Second, it gives an initial outer weight that guarantees a single-atom insertion decrease before the local semismooth Newton correction.

For a nonhomogeneous activation set $R_{\text{search}}(\mu_t) := \min\{R(\mu_t), e^5\}$. Each multistart initial point has norm at most $R_{\text{search}}(\mu_t)$. Starting from each ω_0 , one unconstrained L-BFGS solve jointly optimizes all components of ω for $|P_{p,\mu_t}^M(\omega)|$. After optimization, candidates with $|\omega| > R_{\text{search}}(\mu_t)$ are discarded. This avoids constrained optimization.

Before running the SSN solver, the algorithm calculates

$$(58) \quad A_p^M(\omega) := \left\| \frac{K(\omega)}{w_p(\omega)} \right\|_M^2, \quad \kappa_A(\omega) := -\frac{\Delta(\mu_t, \omega)}{A_p^M(\omega)} \text{sign}(P_{p,\mu_t}^M(\omega)), \quad c_A(\omega) := \frac{\kappa_A(\omega)}{w_p(\omega)}.$$

The coefficient $c_A(\omega)$ is used to initialize the subsequent semismooth Newton correction, which optimizes all outer weights. The decrease statement holds for every accepted candidate, not only for a global maximizer of $|P_{p,\mu_t}^M|$.

Writing $h_\gamma(z) := \phi_\gamma(|z|) - |z|$ and $\tau = (1 + \alpha\gamma)^{-1}$, the semismooth Newton solve uses the coordinatewise residual

$$(59) \quad u = \text{prox}_{\tau\alpha|\cdot|} \left(u - \tau \left[\nabla_u l^M \left(\sum_{n=1}^N \frac{u_n}{w_p(\omega_n)} \delta_{\omega_n} \right) + \alpha \nabla h_\gamma(u) \right] \right).$$

After the solve, the physical outer weights are recovered as $c_n = u_n/w_p(\omega_n)$; the proximal map in (59) is ordinary soft thresholding.

Algorithm 1 uses the scalar penalty ϕ_γ from (32) and the objective (56), with the insertion condition (57).

Algorithm 1 Iterative node insertion for J^M

- 1: Choose T_{out} , N_{trial} , N_{ins} , $\varepsilon_{\text{prune}}$, and an initial network $\omega^{(0)}, c^{(0)}$ of width $N(0)$; set $t = 0$.
 - 2: **while** $t < T_{\text{out}}$ **do**
 - 3: Set $R_{\text{search}} = \min\{R(\mu_t), e^5\}$. Sample N_{trial} points ω_0 with norm bounded by R_{search} .
 - 4: From each ω_0 , run one unconstrained joint L-BFGS maximization of $|P_{p,\mu_t}^M(\omega)|$.
 - 5: Discard candidates with $|\omega| > R_{\text{search}}$, then remove near-duplicates.
 - 6: Keep the candidates satisfying (57); terminate if there is none.
 - 7: Insert at most N_{ins} nodes with the largest $\Delta(\mu_t, \omega)$, warm-start outer weight by (58).
 - 8: Optimize the outer weights of the enlarged network by the semismooth Newton method.
If the correction increases J^M , discard it and keep the post-insertion weights.
 - 9: Remove nodes with $|c_n| \leq \varepsilon_{\text{prune}}$ (resulting in width $N(t+1)$).
 - 10: $t = t + 1$.
 - 11: **end while**
-

5.2. Algorithm 2: the fractional-exponent penalty on the sphere. Algorithm 2 uses $\psi_k(z) = z^q$, $q = 2/(k+1)$, on the unit sphere. For $q < 1$, insertion condition of Algorithm 1 is unavailable (Remark 4.8). Furthermore, since the zero network is a local minimizer and the infinitesimal, we need good initialization method, which is provided by the Theorem 4.7

We therefore test the actual change produced by inserting one atom with a finite coefficient. We use the scalar proximal map

$$(60) \quad \text{prox}_{\lambda|\cdot|^q}(v) \in \arg \min_{z \in \mathbb{R}} \left\{ \frac{1}{2}(z - v)^2 + \lambda|z|^q \right\},$$

with zero selected when the minimum is attained both at zero and at a nonzero point. For $0 < q < 1$, put

$$\bar{z} = [2\lambda(1 - q)]^{1/(2-q)}, \quad \bar{v} = \frac{2 - q}{2(1 - q)} \bar{z}.$$

Then the selected proximal value is zero when $|v| \leq \bar{v}$; when $|v| > \bar{v}$, it is $\text{sign}(v)z$, where z is the largest positive solution of

$$z + \lambda q z^{q-1} = |v|.$$

The implementation evaluates this value in closed form for $q \in \{1/2, 2/3\}$, using respectively a cubic in $\sqrt{z}$ and a quartic in $z^{1/3}$. At the endpoint $q = 1$, (60) is ordinary soft thresholding.

For a candidate $\omega \notin \text{atom } \mu_t$, write

$$P := P_{\mu_t}^M(\omega), \quad A := \|K(\omega)\|_M^2.$$

If A or $|P|$ is numerically zero, the candidate is discarded. Otherwise the exact one-atom objective increment is

$$(61) \quad \Delta J_{\psi_k}^M(c; \omega) = cP + \frac{1}{2}Ac^2 + \alpha|c|^q.$$

Completing the square in (61) gives its selected global minimizer directly:

$$(62) \quad c^*(\omega) := \text{prox}_{(\alpha/A)|\cdot|^q} \left(-\frac{P}{A} \right).$$

We accept the candidate precisely when $c^*(\omega) \neq 0$ and $\Delta J_{\psi_k}^M(c^*(\omega); \omega) < 0$. Thus the same coefficient both certifies the finite-step decrease and initializes the subsequent correction.

After insertion, let m_t be the smallest magnitude among the nonzero outer weights. For $q < 1$, the correction fixes the proximal scale

$$(63) \quad \lambda_t := \min \left\{ \alpha, \frac{\rho_{\text{prox}} m_t^{2-q}}{2(1 - q)} \right\}, \quad \rho_{\text{prox}} = 0.5,$$

for the whole semismooth Newton solve. Every nonzero warm-start coefficient is strictly above the switching magnitude, since $\bar{z} \leq \rho_{\text{prox}}^{1/(2-q)} m_t < m_t$. Define, coordinatewise,

$$(64) \quad c(z) := \text{prox}_{\lambda_t|\cdot|^q}(z), \quad G_t(z) := \frac{\alpha}{\lambda_t}(z - c(z)) + \nabla_c l^M(\mu_{\omega, c(z)}).$$

The coefficient correction applies semismooth Newton to $G_t(z) = 0$. For $q = 1$ we use soft thresholding with $\lambda_t = \alpha$. For $q < 1$, an exact undamped Newton iteration on a fixed smooth branch has superlinear convergence rate locally under nonsingularity.

The implementation uses $k \in \{2, 3\}$, corresponding to $q \in \{2/3, 1/2\}$. The ReLU- ℓ^1 baseline reported alongside it is not this algorithm at $k = 1$: it is the classical profile-threshold insertion with a coordinate-descent warm start and the convex penalty $\alpha \sum_i |c_i|$, run separately.

Algorithm 2 Iterative node insertion for $J_{\psi_k}^M$

-
- 1: Choose T_{out} , N_{trial} , N_{ins} , $\varepsilon_{\text{prune}}$, and an initial network $\omega^{(0)}, c^{(0)}$ of width $N(0)$; set $t = 0$.
 - 2: **while** $t < T_{\text{out}}$ **do**
 - 3: Sample N_{trial} nodes randomly as the initial points.
 - 4: For each initial point, run unconstrained L-BFGS w.r.t. $|P|$ and normalize onto $\mathbb{S}^d$.
 - 5: Remove cosine near-duplicates and discard final candidates that coincide numerically with the current support.
 - 6: For each candidate compute $c^*(\omega)$ from (62), and keep it when $c^*(\omega) \neq 0$ and $\Delta J_{\psi_k}^M(c^*(\omega); \omega) < 0$; terminate if there is none.
 - 7: Retain at most N_{ins} candidates with the smallest $\Delta J_{\psi_k}^M(c^*; \omega)$, insert with outer weight c^* .
 - 8: Apply the semismooth Newton correction (64), using (63). If the correction increases $J_{\psi_k}^M$, discard it and keep the inserted coefficients c^* .
 - 9: Remove nodes with $|c_n| \leq \varepsilon_{\text{prune}}$ (resulting in width $N(t+1)$).
 - 10: $t = t + 1$.
 - 11: **end while**
-

6. NUMERICAL EXAMPLES

The experiments use seed 42. Each reported parameter choice is run once, and the validation sets are random subsets of the stored sample rows, with 90% used for training and 10% for validation. Input and value normalization are computed from the full sample before the split.

For a fitted value function $\widehat{V}$, the relative errors are

$$\begin{aligned} \text{Err}_{L^2} &:= \frac{\|\widehat{V} - V\|_{L^2}}{\|V\|_{L^2}}, \\ \text{Err}_{H^1} &:= \frac{(\|\widehat{V} - V\|_{L^2}^2 + \|\nabla \widehat{V} - \nabla V\|_{L^2}^2)^{1/2}}{(\|V\|_{L^2}^2 + \|\nabla V\|_{L^2}^2)^{1/2}}. \end{aligned}$$

Both norms are computed empirically on the validation or evaluation set specified for each experiment with the original scale before normalization. For a frontier, support sizes are sorted and the reported error is the smallest among recorded iterates with no more atoms; it is not an iteration trace.

6.1. Van der Pol Oscillator. We consider the optimal control of the Van der Pol oscillator:

$$\min_{u \in L^2(0, T; \mathbb{R})} \mathcal{C}(u; 0, x) = \int_0^T (y_1^2(t) + y_2^2(t) + \eta u^2(t)) dt$$

subject to

$$\begin{cases} \frac{dy_1}{dt} = y_2, \\ \frac{dy_2}{dt} = -y_1 + y_2(1 - y_1^2) + u, \\ (y_1(0), y_2(0)) = (x_1, x_2), \end{cases}$$

In this setting, $d = 2$, $m_u = 1$, $\eta = 0.1$, $T = 3$, and

$$f(y) = \begin{pmatrix} y_2 \\ -y_1 + y_2(1 - y_1^2) \end{pmatrix}, \quad g(y) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad h(y) = y_1^2 + y_2^2.$$

By the optimality condition, $u^* = -\frac{1}{2\eta}g(y^*)^\top p^* = -\frac{1}{2\eta}p_2^*$. The two-point boundary value problem (TPBVP) becomes

$$(65) \quad \begin{cases} \dot{y}_1^* = y_2^*, \\ \dot{y}_2^* = -y_1^* + y_2^*(1 - y_1^{*2}) + u^*, \\ -\dot{p}_1^* = 2y_1^* - p_2^*(1 + 2y_1^*y_2^*), \\ -\dot{p}_2^* = 2y_2^* + p_1^* + p_2^*(1 - y_1^{*2}), \\ y^*(0) = x, \quad p^*(T) = 0. \end{cases}$$

We solve (65) by adaptive collocation using SciPy’s `solve_bvp`, with a residual tolerance of 10^{-8} . The resulting dataset contains 900 value–gradient samples on a 30×30 grid over $[-3, 3]^2$, split into 810 training and 90 validation samples.

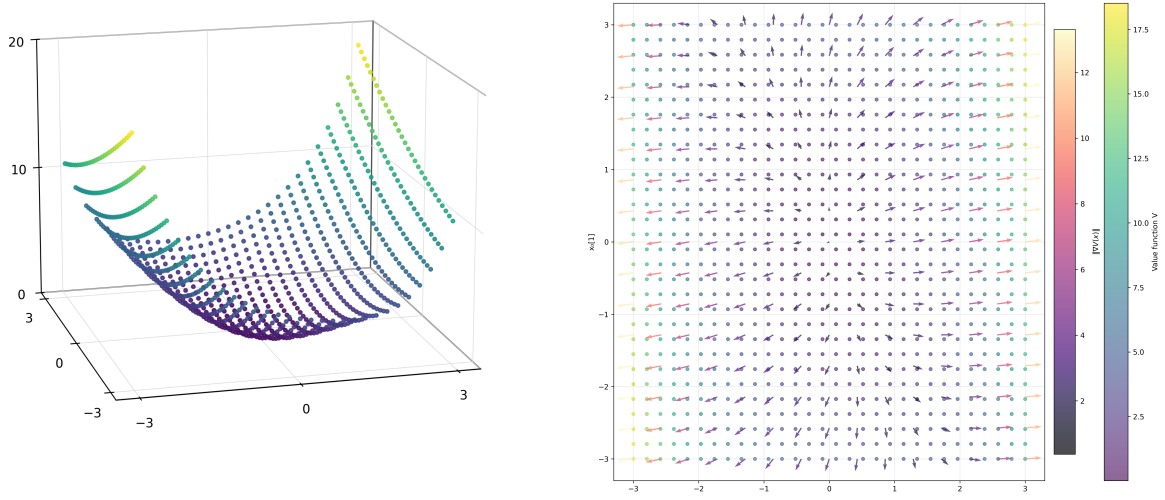


FIGURE 2. Van der Pol numerical value and gradient samples over $[-3, 3]^2$. Left: value labels v^m . Right: v^m (dot color) and costate labels g^m (arrows); arrow color indicates $|g^m|$.

The Algorithm 1, Algorithm 2 use at most $T_{\text{out}} = 150$ outer iterations and $N_{\text{trial}} = 50$ sampled candidate nodes per iteration. Each iteration inserts the single selected candidate ($N_{\text{ins}} = 1$), so the one-candidate decrease statement applies. Corollary 5.2 remains conditional on the selected candidate being an exact global maximizer and is not claimed for the multistart search. The loop terminates when no candidate clears the insertion threshold. The separate traditional ReLU– ℓ^1 comparators use 10 outer iterations with batch insertion of at most 15 candidates per iteration, from 50 candidate starts. Their insertion schedule therefore differs from the sequential nonconvex fits.

Algorithm 1: the log penalty on the normalized measure. The moment order enters the objective through the weight $w_p(\omega) = 1 + |\omega|^p$ inside the scalar penalty: an atom at ω is charged $\phi_\gamma(w_p(\omega)|c|)$ rather than $\phi_\gamma(|c|)$, so a fixed coefficient becomes more expensive the further the atom sits from the origin. The exponent p is therefore an active parameter of the objective.

We use Algorithm 1 with $\gamma = 1$, equal value and gradient weights, and seed 42, for softplus, tanh, the Gaussian and GELU², over

$$\alpha \in \{10^{-4}, 10^{-5}\}, \quad p \in \{2.01, 2.5, 3, 4\}.$$

1. The computable search radius: Theorem 5.1 supplies the radius (51) outside which no candidate can satisfy the insertion condition. Its constants are available before the fit: the growth exponents s_0, s_1 and the constant C_ρ of Assumption 3.1 are properties of the activation, and the remaining factor follows from the sample extent, so $C = C_\rho(A_M^{s_0} + A_M^{s_1-1})$ with $A_M = \max\{1, \max_m \sqrt{1 + |x^m|^2}\}$. The implementation also imposes the fixed bound $\lambda \leq e^5$, so the theorem radius changes the search only when it is smaller than that bound.

The recorded search-radius histories fall below e^5 in thirteen of the thirty-two Van der Pol moment-order runs: one at $p = 2.5$, five at $p = 3$, and seven at $p = 4$. The $p = 2.5$ case is the Gaussian at $\alpha = 10^{-4}$. At $p = 3$, both tanh and the Gaussian activate the smaller cap at both values of α , as does softplus at $\alpha = 10^{-4}$. At $p = 4$, all three activate it at both values of α , together with GELU² at $\alpha = 10^{-4}$.

We investigate the effect of including gradient data for softplus, tanh, and the Gaussian. Table 1 compares fitting the values alone with fitting both values and gradients.

2. Effect of the gradient training: With the H^1 loss, all three relative gradient errors fall substantially: from 0.328 to 0.0453 for softplus, from 0.358 to 0.0327 for tanh, and from 0.273 to 0.0169 for the Gaussian. Value accuracy improves as well (Table 1). The Gaussian reaches relative H^1 error 0.0111 with 32 atoms; tanh reaches 0.0218 with 29, and softplus 0.0311 with 14.

TABLE 1. Effect of gradient-augmented fitting with $\alpha = 10^{-4}$, $\gamma = 10$, $p = 2.01$, seed 42. The same parameters are used for all three activations, so the comparison is not confounded with per-activation tuning.

Activation	L^2 trained	H^1 trained	Activation	L^2 trained	H^1 trained
softplus	8.38×10^{-2}	1.24×10^{-2}	softplus	2.23×10^{-1}	3.11×10^{-2}
tanh	7.59×10^{-2}	5.85×10^{-3}	tanh	2.41×10^{-1}	2.18×10^{-2}
Gaussian	5.28×10^{-2}	2.09×10^{-3}	Gaussian	1.82×10^{-1}	1.11×10^{-2}
(A) Effect of gradient training on Err_{L^2}			(B) Effect of gradient training on Err_{H^1}		

3. Choice of activation function: We compare softplus, tanh, and the Gaussian. Because the gradient-augmented loss contains both the value and gradient errors, $K(\omega)$ contributes to the approximation of V and $\nabla K(\omega)$ contributes to the approximation of ∇V . Figure 3 contrasts the three activations through their value, first derivative, and second derivative. The softplus derivative increases from 0 to 1, the derivative of tanh tends to zero at both ends of the real line, and the Gaussian derivative is localized and changes sign. These derivatives generate different columns $a\rho'(a \cdot x + b)$ in the gradient fitting problem.

For the tested parameters, the Gaussian has the smallest relative H^1 validation error, while softplus uses the fewest atoms. Thus accuracy and support size give different rankings.

TABLE 2. Support size and validation error for the parameter values in Table 1.

Activation	N	Err_{L^2}	Activation	N	Err_{H^1}
softplus	6	8.38×10^{-2}	softplus	14	3.11×10^{-2}
tanh	10	7.59×10^{-2}	tanh	29	2.18×10^{-2}
Gaussian	10	5.28×10^{-2}	Gaussian	32	1.11×10^{-2}
(A) L^2 trained			(B) H^1 trained		

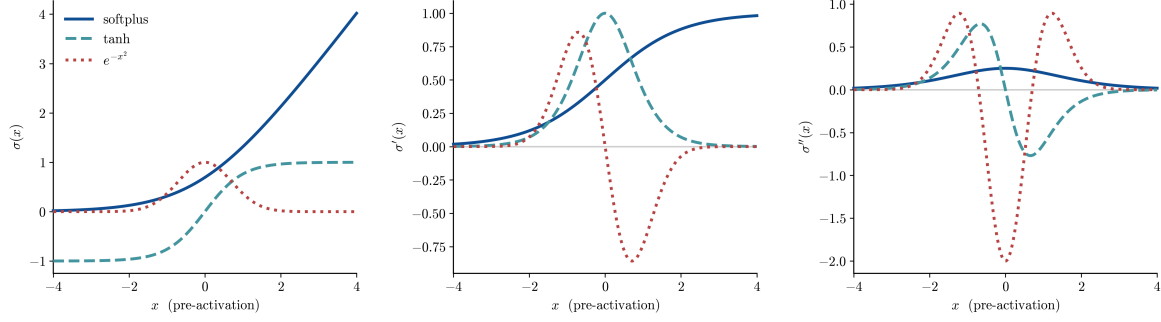


FIGURE 3. The three smooth profiles used by Algorithm 1 — softplus, tanh, and the Gaussian e^{-x^2} — shown as ρ , ρ' , and ρ'' . Under the gradient-augmented loss, the first derivative is the basis that reconstructs ∇V .

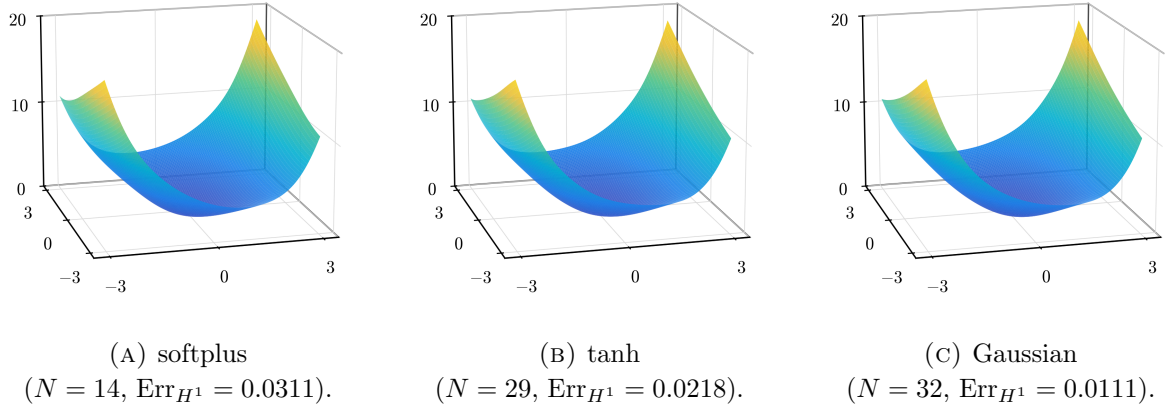


FIGURE 4. Learned value functions under the H^1 loss using the parameter values in Table 1. All three fit the numerical reference data; they differ in support size and validation error. Predictions are clipped to $[0, 20]$ for visualization; the reported errors use the unclipped network outputs.

Algorithm 2: $|c|^q$ -penalty with ReLU^k activation. Algorithm 2 uses the profile $\rho(z) = \text{ReLU}(z)^k$ with the penalty ψ_k , namely $\alpha \sum_i |c_i|^q$ with $q = 2/(k+1)$. The swept parameter is the power k ; all fits below use $\alpha = 10^{-5}$.

For these fixed- α , single-seed fits, adding gradient data lowers the combined value–gradient error from 0.0813 to 0.0114 for $k = 2$, and from 0.0875 to 0.0140 for $k = 3$, factors of about 7.13 and 6.26, respectively (Table 3). Value error also decreases, while the support becomes larger. This comparison alone does not establish better closed-loop performance.

TABLE 3. Algorithm 2: network size and validation error vs. the atom power k ($\alpha = 10^{-5}$).

Power	N	Err_{L^2}	Err_{H^1}	Power	N	Err_{L^2}	Err_{H^1}
$k = 2$	14	2.55×10^{-2}	8.13×10^{-2}	$k = 2$	39	2.75×10^{-3}	1.14×10^{-2}
$k = 3$	13	2.79×10^{-2}	8.75×10^{-2}	$k = 3$	24	5.24×10^{-3}	1.40×10^{-2}
(A) L^2 trained				(B) H^1 trained			

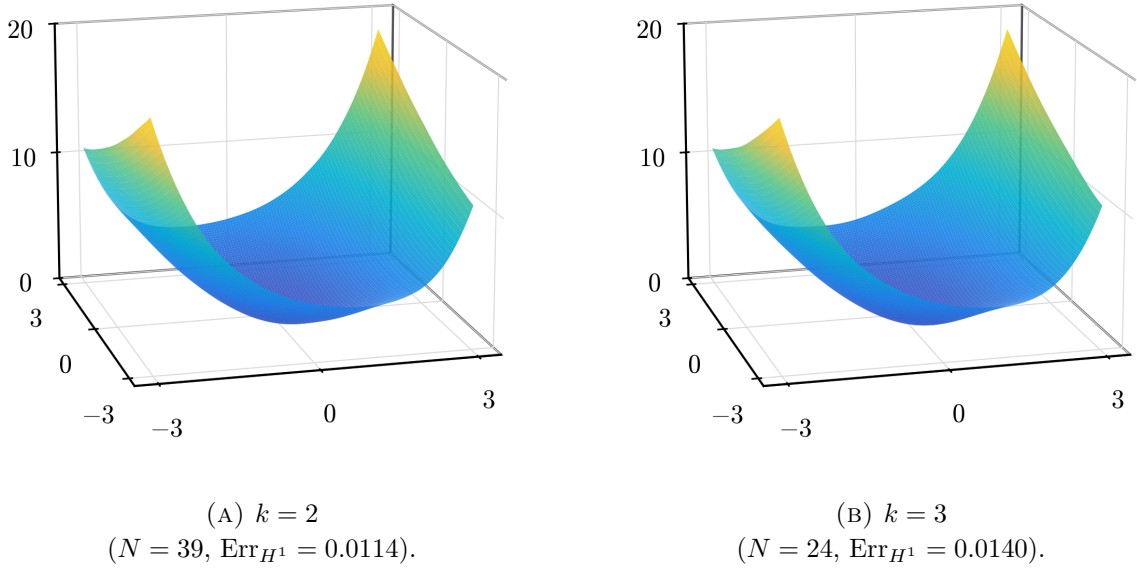


FIGURE 5. H^1 -trained networks with $\alpha = 10^{-5}$ for $k \in \{2, 3\}$. Predictions are clipped to $[0, 20]$ for visualization; the reported errors use the unclipped network outputs.

Changing k changes both the activation and the induced exponent $q = 2/(k+1)$. At comparable H^1 error, the $k = 3$ fit uses 24 atoms instead of 39 for $k = 2$. Under the L^2 loss the fits use 14 and 13 atoms, respectively, with relative H^1 errors 0.0813 and 0.0875 (Table 3(a)).

Figure 6 plots the best observed relative H^1 error at each support budget along the selected fits. At the shared Algorithm 1 operating point, softplus uses 14 atoms at error 0.0311 and the Gaussian uses 32 at 0.0111. At $\alpha = 10^{-6}$, the selected minimum-training-objective Algorithm 2 checkpoints use 70 atoms at 0.00559 for ReLU^2 and 30 atoms at 0.00860 for ReLU^3 .

Feedback law. We simulate trajectories under the explicit learned feedback laws from $y_0 = (2, 1)$ over $T = 3$, using fixed-step RK4 with $\Delta t = 0.01$, zero-order-hold controls, and the numerical bound $|u| \leq 50$. Figure 7 compares the signed control, state norm, and accumulated running cost for all five fitted controllers with a numerical open-loop reference. The learned laws have running costs between 6.415 and 6.449, compared with 6.401 for the numerical open-loop reference.

The compared networks have 29 neurons for tanh, 14 for softplus, 32 for the Gaussian, 70 for ReLU^2 , and 30 for ReLU^3 . The nonhomogeneous fits use $(\alpha, \gamma, p) = (10^{-4}, 10, 2.01)$, and both homogeneous fits use $\alpha = 10^{-6}$. The same checkpoints supply the summary widths above.

Figure 8 shows the inner parameters for the Gaussian, softplus, and ReLU^k ($k = 3$). The two algorithms produce structurally different representations: Algorithm 2 constrains its atoms to the unit sphere $\mathbb{S}^2$, while Algorithm 1 leaves them free — the Gaussian atoms in particular span a wide range of norms.

6.2. Pendulum Swing-Up. The question in this example is how a sparse gradient-trained network approximates a continuous value function whose gradient jumps across a switching set. We observe the pendulum example: we write the state as

$$x = (x_1, x_2)^\top = (\theta, \dot{\theta})^\top,$$

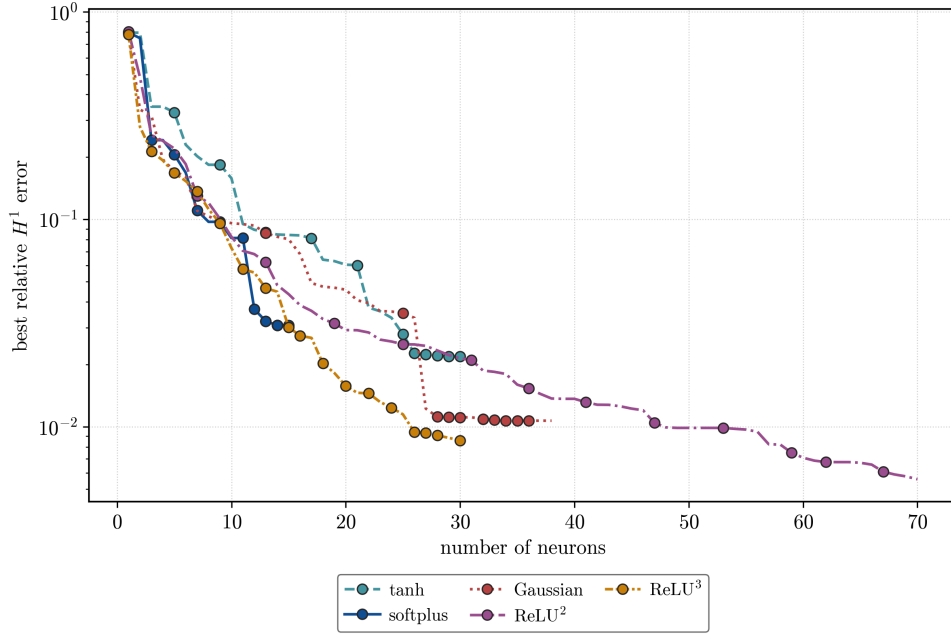


FIGURE 6. Best observed relative H^1 validation error against support budget. Nonhomogeneous profiles use the parameter values stated in Table 1; homogeneous profiles use $\alpha\psi_k$ at $\alpha = 10^{-6}$ for both $k = 2$ and $k = 3$.

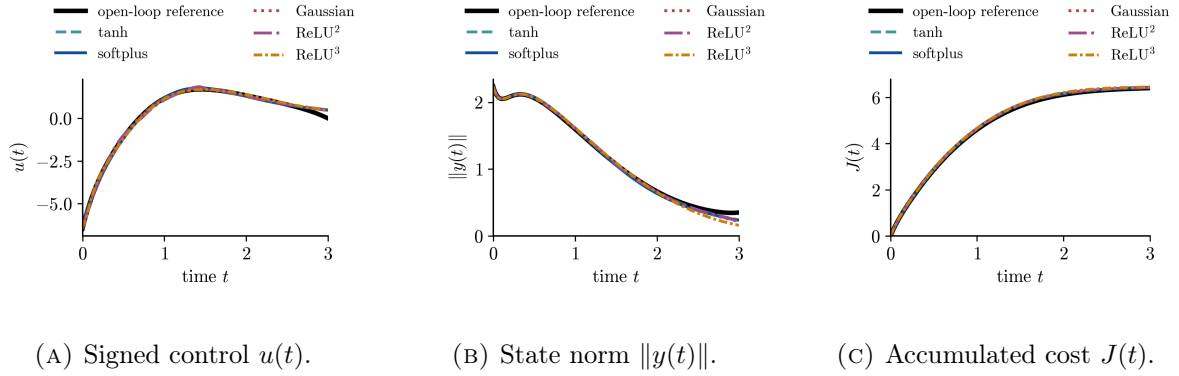


FIGURE 7. Van der Pol controls, state norms, and realized costs from $y_0 = (2, 1)$ over $T = 3$. All three panels show the same five learned controllers and numerical open-loop reference (black). The accumulated cost $J(t) = \int_0^t \ell(y(s), u(s)) ds$ uses the held-control RK4 stages for learned laws and three-point Gauss quadrature on the dense PMP solution for the reference, without a learned terminal-value contribution.

the pendulum angle and angular velocity. We consider the optimal control problem

$$\min_{u \in L^2(0, \infty; \mathbb{R})} \mathcal{C}(u; 0, x) = \int_0^\infty (\sin^2 \theta(t) + (\cos \theta(t) - 1)^2 + \dot{\theta}^2(t) + \eta u^2(t)) dt$$

subject to

$$\begin{cases} m\ell^2 \ddot{\theta} = -b\dot{\theta} + mg\ell \sin \theta + u, \\ (\theta(0), \dot{\theta}(0)) = (x_1, x_2). \end{cases}$$

The target equilibria are $x_{U,j} := (2j\pi, 0)$, $j \in \mathbb{Z}$; for a selected target we write $x_U = x_{U,j}$. Here $m = 1$ is the pendulum mass, $\ell = 1$ the pole length, $b = 0.1$ the damping coefficient, $g = 9.8$ the

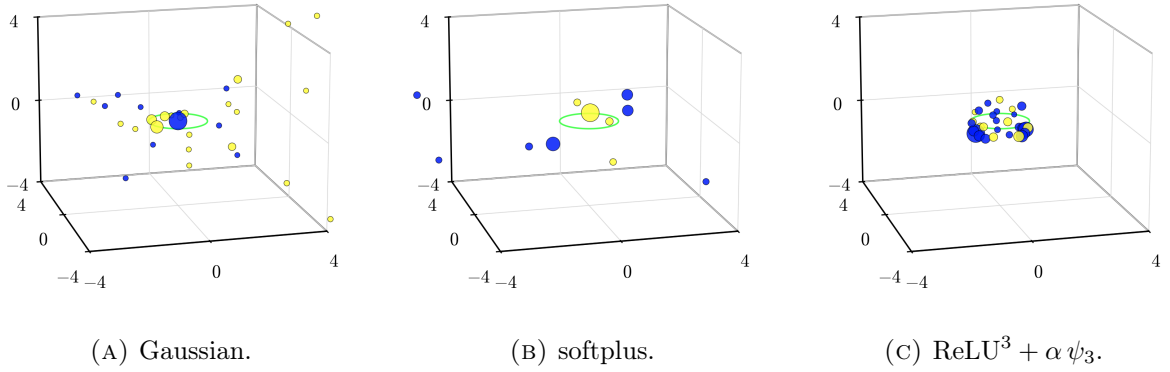


FIGURE 8. Inner parameters (a_1, a_2, b) , with the unit circle in the plane $b = 0$ shown in light green. The two nonhomogeneous fits use $\alpha = 10^{-4}$, $\gamma = 10$, $p = 2.01$; dot color gives the sign and dot size the magnitude of the outer weight.

gravitational acceleration, and $\eta = 1$ the control-cost weight (denoted r in [10]); u is the applied torque.

The open loop algorithm follows from [10]. The state-value-gradient triple is generated with discontinuous gradient.

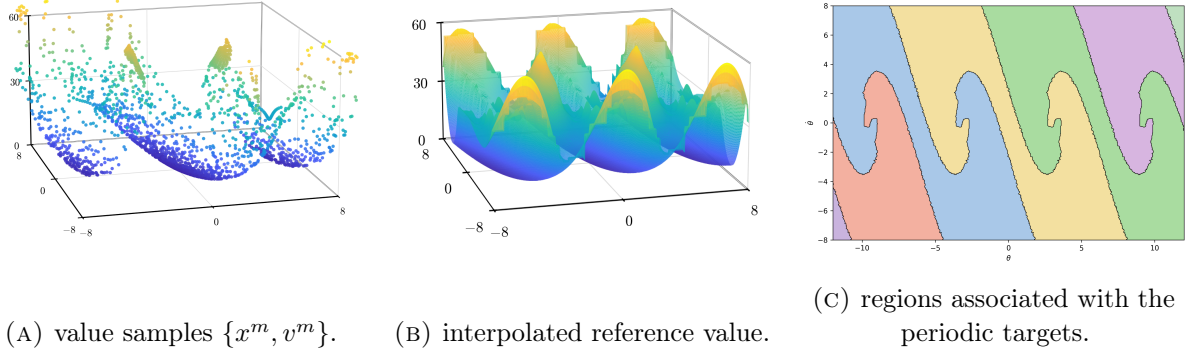


FIGURE 9. Numerical pendulum reference data over the $(\theta, \dot{\theta})$ plane. (a) Selected value samples from the PMP generator. (b) A periodic interpolation of those samples. (c) Target-region visualization obtained by assigning each grid point to the nearest sampled first-exit characteristic and its periodic copies. This coloring approximates the target regions; it does not compare branch costs at every grid point.

Approximation. For each model family, we first retain parameter combinations that satisfy the numerical stabilization criterion below from both A and B at $T = 40$, then select the smallest global relative H^1 validation error among them. Each run contributes its minimum-training-objective checkpoint. If no combination passes, we show the lowest-global-error fit as a failed comparator. Thus A and B are selection states; their reported outcomes are not an independent evaluation of control generalization. We consider the existing H^1 -trained fits on the same production dataset: 24 runs per nonhomogeneous activation across the log-penalty and moment-order sweeps, four per rectified power, and five for the $\text{ReLU}-\ell^1$ baseline. Gaussian and ReLU^3 have no passing candidate; softplus, tanh, ReLU^2 , and the baseline do.

Figure 10 tracks the best observed relative H^1 validation error at each support budget within each selected run. At the selected checkpoints the global errors are 0.200 for ReLU^2 , 0.209 for the Gaussian, 0.232 for $\tanh$, 0.248 for ReLU^3 , and 0.274 for softplus.

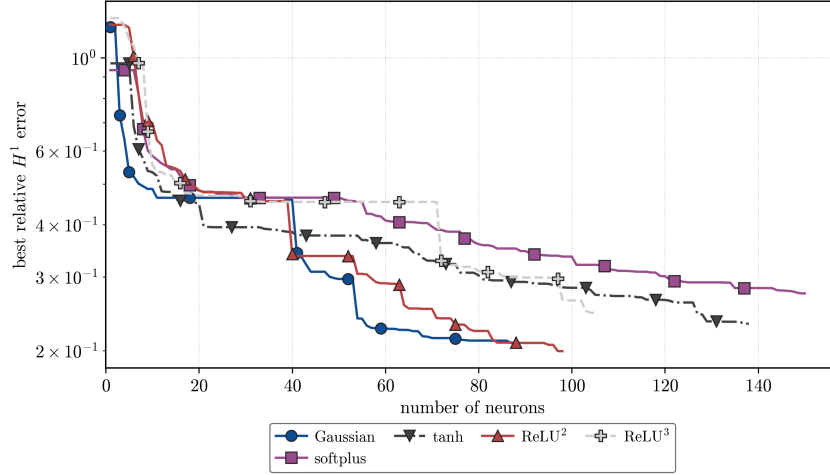


FIGURE 10. Best relative H^1 validation error against neuron count. The nonhomogeneous activations use (56); the homogeneous activations use $J_{\psi_k}^M$. All curves use the selected runs and parameters in Table 4.

Figure 11 shows the fitted surfaces on the same physical window and value range as the value function in Figure 9b. The network shows limitation in approximating the optimal value function.

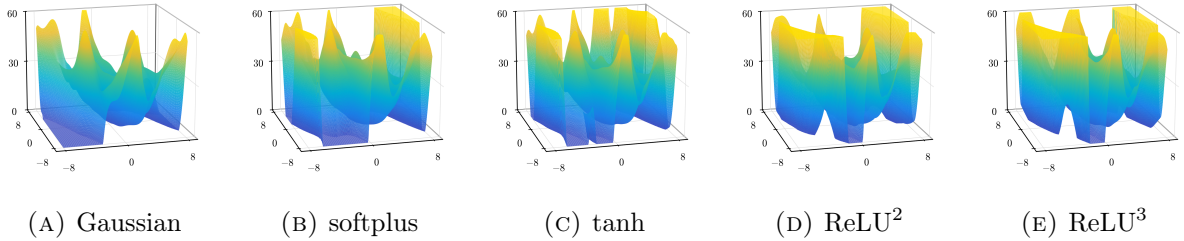


FIGURE 11. Learned value surfaces $\hat{V}(x)$ over the $(\theta, \dot{\theta})$ window $[-8, 8]^2$. Predictions below 0 or above 60 are clipped to the respective boundary for visualization; errors and feedback use the unclipped fitted function. The ReLU^k models are trained with the fractional-exponent penalty ψ_k , and the nonhomogeneous models with (56) using the parameter values reported in Figure 1b.

Feedback law. Minimizing the Hamiltonian in u gives the optimal control in feedback form,

$$u(x) = -\frac{1}{2\eta m\ell^2} \partial_{\dot{\theta}} V(x);$$

replacing V by a fitted $\hat{V}$ yields a candidate feedback law. We test each law in closed loop from two initial states placed on either side of the switching curve, $A = (0.74, 0.61)$ and $B = (0.26, 0.46)$, integrating the dynamics up to $T = 40$ with fixed-step RK4 ($\Delta t = 5 \times 10^{-3}$), zero-order-hold (ZOH) controls, and the numerical bound $|u| \leq 30$.

We assess *numerical stabilization* using the following criterion: at every recorded sample in $[T - 2, T]$, the trajectory must satisfy

$$\text{dist}(\theta(t), 2\pi\mathbb{Z}) < 0.1, \quad |\dot{\theta}(t)| < 0.5.$$

This finite-horizon criterion does not establish asymptotic stability. We assess horizon sensitivity at $T \in \{10, 20, 40\}$ using the same checkpoints and initial states.

From the harder start A , softplus, tanh, ReLU², and the ReLU- ℓ^1 baseline satisfy the numerical stabilization criterion at all three horizons; Table 4 reports $T = 10$ and $T = 40$ for both starts. From B , all five selected nonconvex fits pass at all three horizons; the baseline passes at $T = 20, 40$ but not at $T = 10$. Halving the integration and control-hold step preserves these classifications.

Feedback law	N	Start A			Start B		
		Stabilized			Stabilized		
		$J(10)$	$T = 10$	$T = 40$	$J(10)$	$T = 10$	$T = 40$
reference (PMP)	—	26.2	yes	yes	10.2	yes	yes
ReLU + ℓ^1	72	82.2	yes	yes	115.7	no	yes
Gaussian	88	361.4	no	no	10.2	yes	yes
softplus	150	307.9	yes	yes	10.1	yes	yes
tanh	138	71.7	yes	yes	10.1	yes	yes
ReLU ² + ψ_2	97	86.9	yes	yes	10.1	yes	yes
ReLU ³ + ψ_3	105	69306.3	no	no	10.1	yes	yes

TABLE 4. Pendulum feedback from starts $A = (0.74, 0.61)$ and $B = (0.26, 0.46)$. N is the number of neurons. The cost $J(10)$ uses held-control RK4 stages over the first ten seconds. “Stabilized” means $\text{dist}(\theta, 2\pi\mathbb{Z}) < 0.1$ and $|\dot{\theta}| < 0.5$ at every recorded sample in $[T - 2, T]$, assessed separately for each column. The reference is the sampled-PMP feedback heuristic. Selection first requires stabilization from both A and B at $T = 40$, then minimizes global validation H^1 error.

Figure 12 compares all five learned controllers with numerical open-loop references from both starts. As in the Van der Pol comparison, the reference for each start is the least-cost converged finite-horizon PMP solve initialized from the complete learned and dataset-feedback rollouts. It uses $T = 40$, a free terminal state, and zero terminal cost.

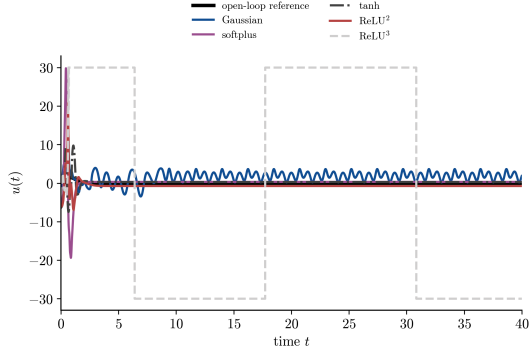
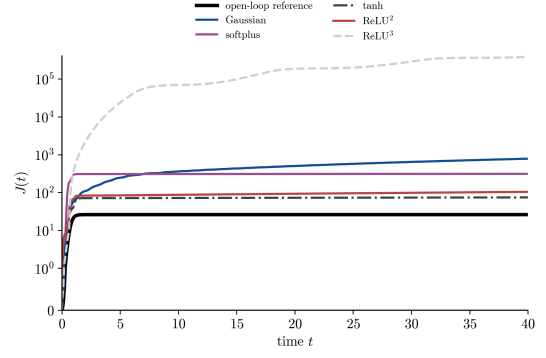
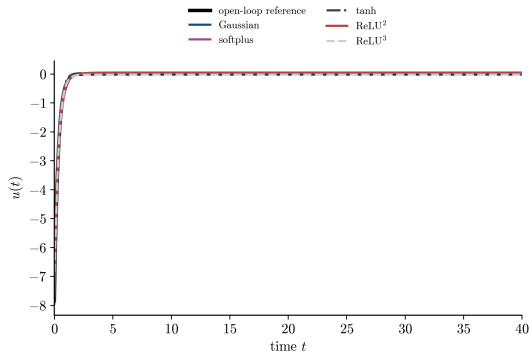
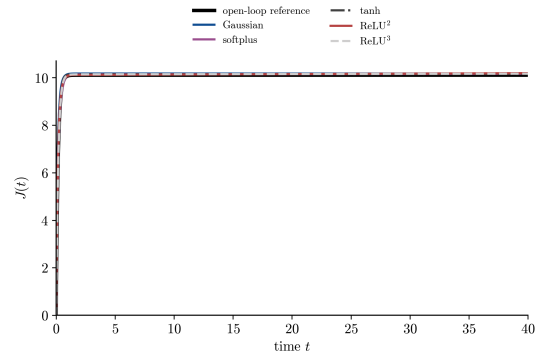
(A) Control from A .(B) Accumulated cost from A .(C) Control from B .(D) Accumulated cost from B .

FIGURE 12. Pendulum controls and realized costs up to $T = 40$. Each learned feedback law is simulated on its own trajectory with $\Delta t = 0.005$ and the numerical control bound $|u| \leq 30$. The black curves are numerical open-loop references with matching initial states and horizons.

Validation errors on gradient discontinuity. Let S denote the boundary across which the value function V remains continuous but its gradient jumps. To study approximation error near this gradient discontinuity, we consider the region within distance 0.3 of S :

$$S_{0.3} := \{x \in \mathbb{R}^2 : \text{dist}(x, S) \leq 0.3\},$$

where dist is Euclidean distance in the $(\theta, \dot{\theta})$ coordinates. Numerically, we use reconstructed points on S and copies shifted by $\pm 2\pi$ in the angle coordinate.

Figure 13 reports the relative H^1 error on a shared evaluation set of 994,558 points. The union of training rows across the production and oversampling datasets is excluded, and for each run we use the recorded iterate with the smallest training objective. Errors are reported on the subsets inside and outside $S_{0.3}$. The ReLU² fit has the smallest errors in both regions (0.284 and 0.155). The corresponding values are 0.290 and 0.175 for the Gaussian, 0.331 and 0.194 for ReLU³, 0.329 and 0.220 for tanh, and 0.363 and 0.250 for softplus. Every selected model has a larger relative error near the gradient discontinuity, inside $S_{0.3}$, than outside this region.

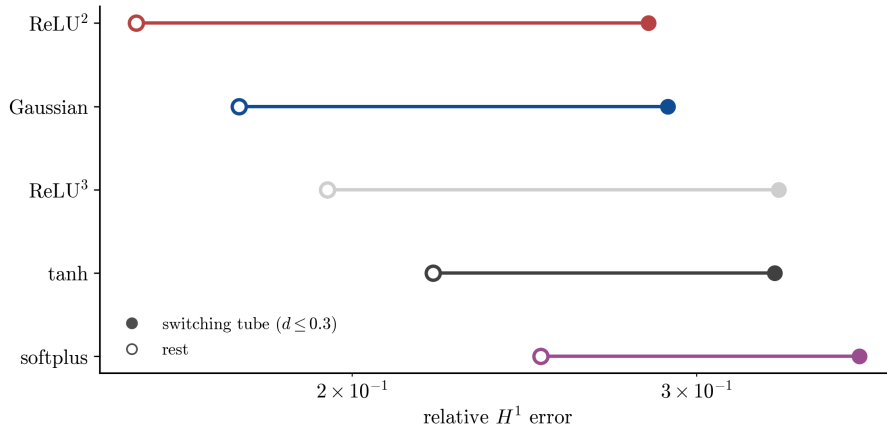


FIGURE 13. Relative H^1 error near the gradient discontinuity (within distance 0.3 of S , filled markers) and on the remaining evaluation points (open markers), with rows ordered by the latter error.

7. CONCLUSION AND OUTLOOK

We developed a measure-theoretic framework for sparse, gradient-augmented recovery of HJB value functions from PMP data. Its main contribution is a variational route from infinitely wide dictionaries to finite shallow networks for two activation classes that require different compactness arguments. For nonhomogeneous activations, simultaneous normalization of the measure and dictionary controls escape on the unbounded parameter domain and, together with the concave penalty, yields existence, finite support, separation and width bounds, and a quantitative decay rate for the excess above the insertion threshold. For positively k -homogeneous activations, radial optimization instead reduces the problem to the unit sphere and induces a fractional-power penalty; the resulting formulation likewise admits global minimizers and explicit finite-width conclusions. These results give a rigorous foundation for constructing sparse networks that approximate both a value function and its gradient.

The numerical studies illustrate the role and the boundary of this framework. On the Van der Pol problem, the learned feedback laws attain running costs close to the numerical open-loop reference over $T = 3$. On the pendulum problem, selecting first for numerical stabilization and then for global validation accuracy identifies passing combinations for softplus, tanh, ReLU²,

and the ReLU- ℓ^1 baseline from both selection states at $T = 40$. No tested Gaussian or ReLU³ combination passes from both states. The selected tanh controller has the lowest cost from the harder start among the nonconvex fits, while ReLU² has the smallest approximation error. Residual running costs remain positive, so the finite-horizon criterion does not establish asymptotic stabilization. Thus gradient-augmented fitting is a useful bridge from PMP data to feedback synthesis, while feedback performance must still be assessed directly when the value gradient is discontinuous.

The most immediate extension concerns the insertion step: the rate analysis assumes exact global candidate maximization, whereas the implementation uses a multistart local search. Developing certified or more reliable candidate searches would therefore bring the algorithm closer to the theoretical guarantee. A second direction is to complement the present approximation criteria with feedback-aware training or validation near switching sets, and then to examine the resulting methods in higher-dimensional control problems.

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